

# SEPARATING CHOICE FROM VALIDITY: A VALIDATED SEARCH-CHOICE MEASUREMENT FOR VISION-LANGUAGE SEARCH POLICIES

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## ABSTRACT

Learned search policies are scored against a random-valid control, uniform over valid operations. On additive best-first cells of our enumeration contract, random-valid makes the exact reference’s decisions on 48/48 pairs, so success measures validity. Submission-order tie-breaking gives 19/48 divergent pairs, though a pre-registered majority prediction failed. Random-valid success measures choice only if its decisions can differ from the reference’s and the budget binds them. In our choice-frontier contract the policy picks the next frontier state, and its solve-versus-budget area M1 passes a pre-registered exact- $\epsilon$  ladder on validation and held-out panels. The privileged rungs validate the measurement, not a policy.

A trained adapter beats random-valid M1 by  $D_3 = +0.285 [+0.140, +0.433]$  over three seeds and by  $+0.416$  on the held-out panel. It is separated above the 0.75 rung only on the held-out panel ( $+0.227 [+0.053, +0.403]$ ), not pooled. On an unscreened panel the gain is  $+0.149 [+0.030, +0.283]$ , resting on 4 of 12 tasks (descriptive). Random-valid registers choice on ScienceWorld (0.915 versus 0). On LLM-First Search, a zero-token chooser saturates token caps by construction but solves 0.0775 at the reference’s expansion count. The 48/48 failure is shown only on our runtime. The choice-quality claim is development-stage (development panels only), with no planning-ability or generality claim.

## 1 INTRODUCTION

A learned search policy is useful only if it chooses well among valid operations, yet its success can also reflect validity alone. On classical planning probes (Valmeekam et al., 2023b; Stein et al., 2025; Kokel et al., 2026), off-the-shelf language models improve at operational reasoning but not at structural enumeration (Huang et al., 2026). We train a Search Process Policy to emit Typed Search Operations, which a no-repair Trusted Search Runtime validates, under text, visual, and multimodal observations (Section 3). The declared algorithms are breadth-first search (BFS), width-based search, and additive best-first search, which is best-first search ordered by the additive heuristic  $h_{\text{add}}$ . The natural control is random-valid, uniform over valid operations, but without a check a benchmark can credit validity as choice.

On part of our own runtime, random-valid makes exactly the decisions of the exact reference, the declared algorithm run as specified. In the enumeration contract’s additive best-first cells, where each expansion submits its complete candidate set and ties break in sorted order, the two agree on 48/48 pairs (Figure 1a). Breaking ties in submission order gives 19/48 divergent pairs, so the identity audit, a decision-by-decision comparison of the two, can return either verdict. A pre-registered prediction of more than 24/48 divergent pairs failed (Section 5.2). Process supervised fine-tuning (SFT) solved 125 of 288 baseline entries, all in these cells, and 0/24 in each BFS and Best-First Width Search (BFWS) cell (Section 5.6).

Random-valid success therefore measures choice only if the identity audit finds divergent decisions and the budget check, a test of whether the success budget binds decisions, finds that it does. We build a choice-frontier contract, where the policy picks the runtime-owned frontier state to expand under a budget of twice the reference’s expansion count (Figure 1b). M1, the area under the solve-versus-budget curve, is first validated by a pre-registered ladder of exact- $\epsilon$  selectors, which random-

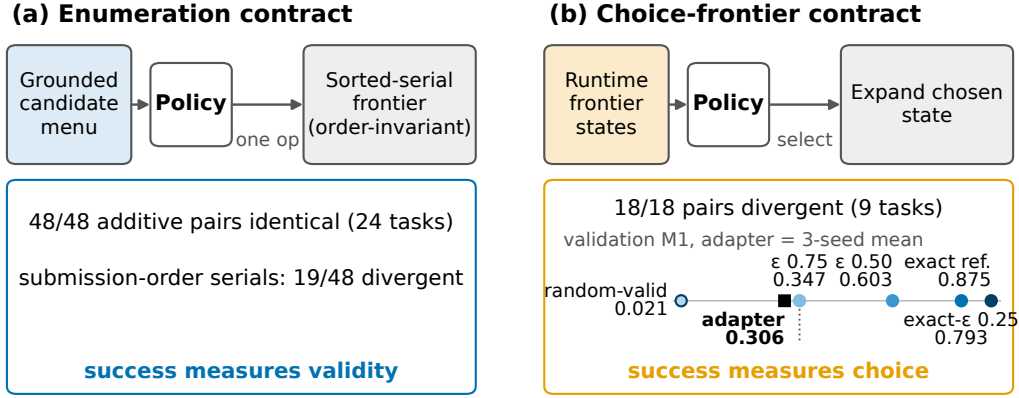


Figure 1: The two decision contracts and what random-valid success measures under each. (a) Under the enumeration contract, random-valid makes the exact reference’s decisions on 48/48 additive pairs, and 19/48 pairs diverge when ties break in submission order. (b) Under the choice-frontier contract, 18/18 pairs on the first panel diverge. On the validation panel, M1 falls from 0.875 for the exact reference through the exact- $\epsilon$  rungs to 0.021 for random-valid. The adapter’s three-seed mean is 0.306. Success therefore measures validity in (a) and choice in (b), and the rungs read privileged information.

ize a fraction  $\epsilon$  of reference decisions. On the validation panel, 12 fresh tasks from 7 domains, the ladder passes. Every adjacent gap is positive (smallest  $+0.082$  [ $+0.051$ ,  $+0.111$ ], Section 5.3). The rungs read the reference’s own priority order, which no policy sees, so the ladder validates the measurement, not a policy.

Scored this way, a trained adapter beats uniform choice on all three panels, but its ladder position depends on the panel. Its gain  $D_3$ , adapter minus random-valid M1 averaged over three training seeds, is  $+0.285$  [ $+0.140$ ,  $+0.433$ ] on the validation panel, with every seed positive, meeting the pre-registered rule. On the held-out panel, frozen before any evaluation episode,  $D_3 = +0.416$  [ $+0.250$ ,  $+0.591$ ]. It is separated above the 0.75 rung on the held-out panel ( $+0.227$  [ $+0.053$ ,  $+0.403$ ]) but not on the validation panel ( $-0.041$  [ $-0.187$ ,  $+0.102$ ]), the unscreened panel, or pooled over 35 tasks ( $+0.058$  [ $-0.040$ ,  $+0.156$ ], descriptive, Section 5.4). On the unscreened panel, of tasks random-valid never solved in screening, the gain is  $+0.149$  [ $+0.030$ ,  $+0.283$ ], resting on 4 of 12 tasks (descriptive). All three gains come from imitating the exact reference.

Both checks matter on published interfaces. On ScienceWorld (Wang et al., 2022), random-valid diverges from the gold trajectory at the first decision in 1000/1000 pairs over 200 tasks. Success is 0.915 for the reference and 0 for random-valid, so it measures choice there. The audit covers 6 of 30 task types, and no surveyed interface uses an enumeration contract, so the 48/48 failure is shown only on our runtime. On LLM-First Search (Herr et al., 2025), a zero-token chooser saturates the token caps by construction, so success must count against a budget that binds decisions (Section 5.5).

This paper makes four contributions.

- **The identity audit, a per-decision check that returns both verdicts on one runtime.**
- **A choice-frontier measurement validated by a pre-registered exact- $\epsilon$  ladder before any policy is scored.**
- **A trained adapter above uniform choice on three panels, with its ladder position and screening boundary.**
- **The budget check, with an audit of two published interfaces, random-valid registering choice on one and saturating by construction on the other.**

Every claim is development-stage, measured on development panels, not on the frozen 45-task final evaluation, which is unexecuted. Held-out means only that panel, drawn from seeds no adapter had been evaluated on (Section 4). Enumeration-contract cells are single-seed except two replication cells (seeds 17/29/71, descriptive  $n=3$ ). We make no planning-ability or generality claim.

## 2 RELATED WORK

We evaluate step-by-step execution of declared algorithms such as A\* (Hart et al., 1968) or BFWS (Lipovetzky & Geffner, 2017). Planning benchmarks (Valmeekam et al., 2023a;b; Stein et al., 2025; Kokel et al., 2026) instead score plans or plan judgments. Trace-trained models (Lehnert et al., 2024; Gandhi et al., 2024) and step imitators (Velićković et al., 2020; Zaremba & Sutskever, 2014) likewise lack typed operations and in-episode validation. Our choice-frontier contract, where the model chooses which frontier state to expand, is a learned node-selection policy (Orseau & Lelis, 2021). The submission-order counterfactual isolates a heap tie-breaking effect of the kind studied for A\* search (Asai & Fukunaga, 2016).

Our model issues the operations that determine exploration, whereas in Tree of Thoughts (Yao et al., 2023) and RAP (Hao et al., 2023) a scaffold or UCT policy owns it. Scratchpads (Nye et al., 2021), per-step labels (Lightman et al., 2024), Visual Sketchpad (Hu et al., 2024), and MVoT (Li et al., 2025) supervise or generate unchecked free-form steps and imagery rather than attributing control to an observation channel. Unlike repair loops (Kambhampati et al., 2024; Madaan et al., 2023) and syntax-only constrained decoding (Scholak et al., 2021; Willard & Louf, 2023), our runtime never repairs and checks algorithmic invariants on every operation.

Our audit asks whether success reflects choice or only validity. Evaluation-validity work warns that common-sense tasks can confound planning with retrieval (Valmeekam et al., 2023a), and embodied suites such as BEHAVIOR-1K broaden task coverage (Li et al., 2024). Multimodal-training research documents VQA models relying on language priors rather than image content (Zhu et al., 2020), and video-language evaluation reports answer-position selection bias (Loginova et al., 2025). Label-position selection bias is documented for LLM multiple-choice selectors (Zheng et al., 2024), and LVLm chain-of-thought audits find visual or format cues can influence answers without appearing in reasoning traces (Balasubramanian et al., 2025). Our menu manipulations are the corresponding diagnostic, and only the unrun fixed-versus-random-position variant would separate surface-following from state grounding. Process supervision supplies intermediate-step feedback (Lightman et al., 2024), but that signal alone does not test whether a grounded action menu makes frontier choices identifiable.

Our random-valid control follows baseline practice, and our audit extends interface checks to individual decisions. Matched-budget random search is the standard architecture-search baseline (Li & Talwalkar, 2020), and text-game work treats valid-action access as an evaluation handicap. Jericho lists world-change detection for identifying valid actions among its optional handicaps, and CALM reports it matches or beats models given ground-truth admissible actions on half of its unseen games (Hausknecht et al., 2020; Yao et al., 2020). ScienceWorld (Wang et al., 2022) likewise exposes a per-step choice among valid actions, so a random-valid control there registers choice. The identity audit checks this property per decision, and no surveyed published interface implements the enumeration contract on which it fails. This paper operationalizes the known admissible-action handicap in search-execution evaluation with a per-decision identity audit that returns both verdicts on one runtime and anchors the validated choice measurement of Section 5.

## 3 SEARCH PROCESS POLICY

A Search Process Policy executes a declared search algorithm one typed operation at a time. An episode fixes  $(x, A, m, B)$ , which are a task  $x$  in a deterministic grounded planning domain, a declared algorithm  $A$  given as input, an observation adapter  $m$ , and a frozen budget  $B$ . At step  $t$  the policy sees an observation  $o_t$ , a text, visual, or multimodal rendering of the decision context. The observation must expose every fact the exact reference uses at step  $t$ . Search Memory (the runtime-held search state defined below) and the candidates stay textual.

Four definitions fix what the policy owns and what the runtime owns. **Typed Search Operation.** The policy emits a type from  $A$ ’s vocabulary plus every exploration-determining operand, and an emission that omits one is invalid. **Search Memory.** This is bounded state held by the runtime (frontier, visited table, BFWS novelty counters), not model state, and  $o_t$  is the policy’s only channel to history. **Algorithm Invariant.** The runtime checks deterministic per-algorithm properties on every operation, namely FIFO order for BFS, novelty and goal-count priority for BFWS (Lipovetzky & Geffner, 2017), and  $g+3h_{\text{add}}$  or  $h_{\text{add}}$  order under a declared tie-break for weighted A\* (Hart et al.,

1968; Pohl, 1970) and greedy best-first search (Bonet & Geffner, 2001). Compliance establishes neither termination, completeness, optimality, nor success. **Trusted Search Runtime.** The runtime only validates, applies, persists, or rejects operations and never chooses omitted operands, reorders candidates, or repairs, so raw emissions replay to the same trace (Operation Schemas appendix, Appendix A, and Algorithm Invariant Definitions appendix).

The runtime supports two decision contracts, and they differ in what the policy chooses. Under the enumeration contract, each expansion submits its complete candidate set, and the runtime’s priority heap orders it. Under the choice-frontier contract, the policy selects which runtime-owned frontier state to expand next, under a budget of twice the exact reference’s expansion count. Figure 1 and Table 25 (Decision Contracts appendix) compare the two contracts.

Process supervision is token-level cross-entropy cloning of the exact reference’s operation on Search-Trace Segments, which are bounded slices of replay-verified exact traces. Optional DAGger (dataset aggregation) (Ross et al., 2011) adds the exact reference’s corrections on visited contexts.

## 4 EXPERIMENTAL DESIGN

Every learned result is read against two oracle-assisted controls, never as an ability of those controls. The exact reference runs the declared algorithm and defines the imitation target. Where a control saturates, the gain is unmeasurable rather than zero. Under the enumeration contract, random-valid is uniform over valid operations, drawing one listed candidate per decision with a seeded generator and no retry or repair. Candidates are successor rows under additive search, non-duplicate evaluated successors at an admissible frontier position under BFWS, and unvisited successors under BFS, else a frontier-retire operation. Under the choice-frontier contract, random-valid is uniform over the offered frontier, so each random-valid number is read under its own contract.

The identity audit compares random-valid with the exact reference decision by decision. It is the choice-frontier contract’s pre-registered pass condition over 18 seed-17 pairs, passing on any divergence without re-runs, and a diagnostic on the enumeration contract’s additive cells.

The choice-frontier panels are frozen before any evaluation and screened against trivial tasks. The first measurement reused a panel of nine low-cost tasks from 8 domains, shared with the corruption suite, native arms, menu manipulation, and comparator zoo, so each design step could adapt to earlier results. Its training corpus was derived on CPU by replaying the frozen exact traces through the new contract and verified expansion-for-expansion against them (58 episodes bit-exact). The ladder test and the adapter instead use the validation panel, 12 fresh tasks from 7 domains frozen before evaluation. Each candidate task receives 10 screening episodes of random-valid. The screen admits it only if random-valid reaches the goal in 1 to 9 of them, so it admits tasks on which near-uniform choice sometimes succeeds.

The held-out panel (P2 in figures and tables) and the unscreened panel (P2u) were frozen from new seeds before any evaluation episode. Held-out means this panel only, built with the validation panel’s rules from seeds on which no adapter had been evaluated. Its seed extension followed a controls-only screen of the first 40 seeds. The unscreened panel skips the screen and draws kept candidates outside the held-out panel. The frozen 45-task final evaluation remains unexecuted.

Each choice-frontier result has a rule frozen before its outcome. The ladder test passes only if all four adjacent paired 95% lower bounds are positive along the exact reference, the exact- $\epsilon$  rungs, and random-valid. Its intervals are task-clustered, resampling whole tasks (seed 133, 10,000 draws). The adapter rule requires adapter minus random-valid M1 to be positive with a materiality margin of 0.05 and sets an equivalence margin of  $\pm 0.05$ . Later protocols, each frozen before its outcomes, cover three seeds, two fresh panels, a zero-shot policy, and a second external audit.

The enumeration-contract study compares process SFT with its controls. Each cell holds the pre-trained base, process SFT, SFT plus two DAGger iterations (Ross et al., 2011), and an exposure-matched continued-SFT control. SFT trains a fresh low-rank adaptation (LoRA) adapter (Hu et al., 2022) for one epoch on 512 replay-verified records. The BFWS gate checkpoint used two epochs on 47,780 records. The backbone is Qwen3-VL-8B-Instruct (Bai et al., 2025) (revision 0c351dd0), with InternVL3.5-8B (Wang et al., 2025) second. All runs share a 336 GPU-hour cap.

The modality matrix evaluates 24 tasks from 12 domains. Under three observations this gives 288 episodes per arm. Budgets count decision calls, charge invalid operations, cap adapters at twice the exact reference’s count, and price the base at one call. An episode ends at its first invalid emission and counts only after independent replay. Contrasts use a paired whole-instance bootstrap (10,000 resamples, seed 1729, Statistics appendix). Transfer uses exact McNemar tests with Holm correction across 36 comparisons.

Exploratory follow-ups on the reused nine-task panel are reported only in the appendix. Each was frozen before its own first outcome but after earlier outcomes on the same tasks. At submission the BFS pilot had closed, the BFWS gate had passed, and every other stage had executed (Table 2). Generalization and second backbone ran at reduced scope only. The BFWS-to-BFS panel overlap is not pinned. Every enumeration-contract learned cell is single-seed except two replication cells (seeds 17/29/71).

## 5 RESULTS

Results follow the thesis, with the two checks first, then the validated measurement, then the policies it scores. Each measurement is kept separate, the controls are oracle-assisted bounds rather than abilities (Section 4), and every finding is development-stage.

### 5.1 GATE RECEIPTS AND EXECUTED SCOPE

Every executed run reconciles with its ledger and independent replay, and the program spent 68.5244 of its 336 GPU-hour cap. All 1,152/1,152 baseline episodes were independently replayed, and follow-up windows sit on separate ledgers that are never summed. Two branches stopped at their v1 cost gates with terminal governed stops (VALID\_STOP) and completed only at reduced scope. Before the program, the BFS pilot stopped because random-valid also succeeded, the BFWS structural gate passed, and the frozen 45-task final evaluation was never accessed (Gate Receipts appendix, Appendix J, Table 2).

### 5.2 ENUMERATION-CONTRACT IDENTITY AUDIT

On the enumeration contract’s additive cells, random-valid never decides differently from the exact reference. The identity audit compared the two decision by decision on every additive cell of the expanded baseline. They were decision-identical on 48/48 pairs, so success there measures operation validity rather than choice. This is a property of this controller, not of additive best-first search in general (Algorithm Invariant Definitions appendix). The BFS and BFWS cells instead register choice, with random-valid at 15/24 and 17/24. The exact reference reaches 24/24 in both (Table 6).

The same audit returns the other verdict when only the tie-break order changes, finding 19/48 divergent pairs when equal-priority ties break in submission order. The pre-registered prediction of divergence on most pairs (more than 24/48) failed. Divergent pairs first differ between decisions 2 and 50 (Counterfactual appendix, Appendix S). The median first divergence is decision 6, and on the other 29 pairs every equal-priority tie breaks the same way under both orders. Under the choice-frontier contract the audit diverges on 18/18 pairs (Figure 1). The audit must therefore be run on each interface before random-valid success is read as choice.

### 5.3 CHOICE-FRONTIER CONTRACT AND INSTRUMENT VALIDATION

The choice-frontier contract passes the identity audit on its first panel. On the reused 9-task panel, random-valid diverged from the exact reference on 9/9 tasks. Its random-valid minus exact-reference contrast is  $-0.644 [-0.856, -0.422]$  (Comparator-Zoo Receipts appendix, Appendix R, Table 12).

On a fresh validation panel, the pre-registered ladder test passes. The panel has 12 tasks from 7 domains. M1 orders the exact reference (0.875) above the three exact- $\epsilon$  rungs in order, with random-valid (0.021) lowest (Figures 1 and 2). Every adjacent paired task-clustered 95% lower bound is positive, and the smallest gap is  $+0.082 [+0.051, +0.111]$ . The Choice-Frontier Validation appendix (Appendix T, Table 17) gives panel construction, every arm’s receipts, and all four paired gaps.

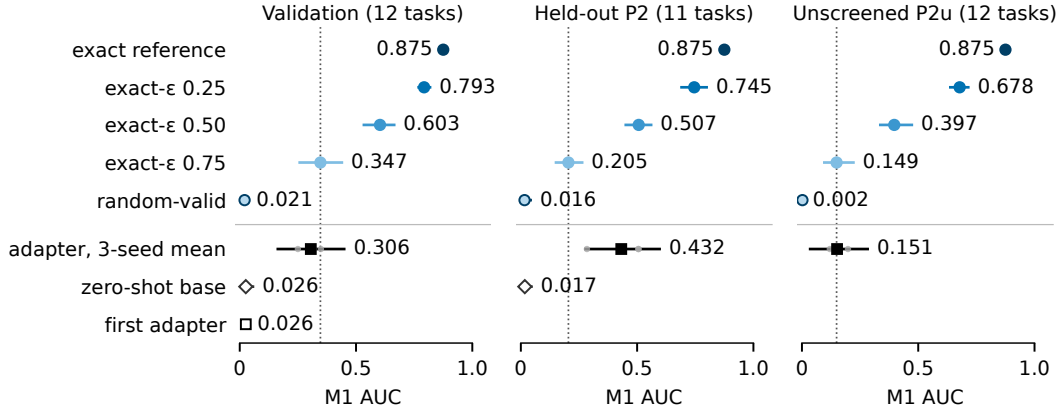


Figure 2: M1, the area under the solve-versus-budget curve, with task-clustered 95% intervals on the validation panel, the held-out panel (P2), and the unscreened panel (P2u). Rows are the five ladder arms, the three-seed adapter mean with faint per-seed points, the zero-shot base (validation and held-out panels), and the first adapter (validation panel only). The rungs fall in order on every panel, the dotted line marks each panel’s 0.75 rung, and only on the held-out panel does the adapter’s interval lie entirely right of it. The rungs read privileged information, so the ladder validates the measurement rather than a policy, and the unscreened ladder is descriptive.

The same test passes on the held-out panel and, descriptively, on the unscreened panel. The held-out panel has 11 tasks from 8 domains, built with the validation panel’s generation, cost, and screening rules on fresh seeds. After the first freeze admitted 6 of the 8 required tasks, an amendment made before any evaluation episode added seeds with the rules unchanged. The added seeds are 955040–955199. On the held-out panel the smallest adjacent gap is  $+0.130$  [ $+0.070, +0.190$ ], and the unscreened pass is descriptive (Held-Out Panels appendix, Appendix U.1). The rungs read the reference’s own priority order, which no policy sees, so the ladder validates the measurement, not a policy.

#### 5.4 POLICIES SCORED BY THE MEASUREMENT

Across three training seeds, the adapter beats uniform choice. On the validation panel  $D_3 = +0.285$  [ $+0.140, +0.433$ ], which meets the pre-registered positive rule. Every seed is positive (Scaled Adapter appendix, Appendix U). On the held-out panel,  $D_3 = +0.416$  [ $+0.250, +0.591$ ]. Seeds 29 and 71 were trained later with the byte-identical recipe under a pre-registered protocol. The first adapter, an earlier small-corpus adapter that differs in corpus, augmentation, sample order, and seed, reaches M1 0.026 [0.010, 0.042] on the validation panel, at the random-valid level.

The adapter’s place on the ladder depends on the panel. It is not separated from the 0.75 rung on the validation panel ( $-0.041$  [ $-0.187, +0.102$ ]), is separated above it on the held-out panel ( $+0.227$  [ $+0.053, +0.403$ ]), and is not separated on the unscreened panel ( $+0.002$  [ $-0.145, +0.157$ ], descriptive). Pooled over 35 tasks it is also not separated ( $+0.058$  [ $-0.040, +0.156$ ], descriptive), and no equivalence margin was pre-registered for this difference, so non-separation is not equivalence. On the validation panel it also falls below the 0.50 rung ( $-0.298$  [ $-0.468, -0.129$ ], descriptive). The held-out separation rests on 11 task clusters, and percentile intervals can under-cover at this size. Per-task differences there are positive on 8 tasks and negative on 3.

A zero-shot policy scores as uniform choice, so the measurement separates the two realisable policies. Prompted zero-shot under the same contract, observation, and cap, the base model is equivalent to uniform choice on the validation panel (development stage),  $D_3 = +0.005$  [ $-0.025, +0.044$ ] within the pre-registered  $\pm 0.05$  margin. On the held-out panel it is inconclusive (confirmatory,  $+0.001$  [ $-0.048, +0.051$ ]). It falls below the 0.75 rung and below the adapter on both panels (Zero-Shot Policy appendix, Appendix U.3). Its contrast with the adapter is  $-0.280$  [ $-0.425, -0.142$ ] on the validation panel. On the held-out panel that contrast is  $-0.415$  [ $-0.589, -0.248$ ].

Without the screen, the gain survives but rests on 4 of 12 tasks. The unscreened panel is drawn from kept candidates outside the held-out panel, and all 12 of its tasks had 0/10 random-valid screen-

ing goals. Both panels were frozen from new seeds before any evaluation episode (Section 4). The ladder passes there (descriptive), and the adapter still beats uniform choice,  $D_3 = +0.149$   $[+0.030, +0.283]$ . The other 8 tasks were unsolved by both the adapter and random-valid (descriptive). This is descriptive evidence that the gain survives without the screen, not evidence about where choice discriminates.

The gain is spread across tasks and domains and comes from imitating the exact reference. Over the three panels the per-task difference is positive on 22 of 35 tasks and zero on 12, with the remaining task negative (descriptive). Leaving out one domain at a time over the validation and held-out panels keeps  $D_3$  from  $+0.305$  to  $+0.395$  (descriptive). Every such lower bound is at least  $+0.185$ . One on-policy DAgger round, run for one seed on half the planned corpus, does not separate the adapter from the 0.75 rung,  $+0.115$   $[-0.026, +0.258]$  over the validation and held-out tasks (DAgger Ablation appendix, Appendix U.2). The choice-quality claim is development-stage, holds relative to uniform choice and the exact- $\epsilon$  ladder on three panels with three training seeds, and carries no planning-ability or generality claim.

## 5.5 EXTERNAL IDENTITY AUDIT

On ScienceWorld (Wang et al., 2022), a published sequential-choice interface, random-valid registers choice. The audit ran five random-valid seeds on 200 tasks under the paper’s 100-step budget. Random-valid diverges from the gold trajectory at the first decision in 1000/1000 pairs. Success is 0.915 for the reference versus 0 for random-valid, so success there measures choice. The audit covers 6 of its 30 task types, and four surveyed candidates were not CPU-feasible under their papers’ own budgets, a judgement call for WebShop and LLM-First Search (Herr et al., 2025). None of the seven published interfaces we surveyed uses an enumeration contract, so the 48/48 identity failure is shown only on our runtime (External Identity Audit appendix, Appendix V).

On LLM-First Search (Herr et al., 2025), random-valid success saturates the paper’s token budget by construction. There the model scores candidates that a search then orders, and the audit covers 400 pairs over 80 tasks from four configurations. The reference model is a local open-weight substitute (Qwen3-30B-A3B-Instruct) for the paper’s GPT-4o, whose published success on these configurations is 0.768 (LLM-First Search Audit appendix, Appendix V.1). Random-valid diverges from the reference by the third decision in every pair, yet under the token budget it solves all tasks (1.0) against 0.825 for the reference. A token cap never binds a chooser that spends no tokens, and because the frontier keeps every alternative, such a chooser is a complete randomized search, here using up to 102,606 expansions against the reference’s 1,300. Counted in expansions, random-valid solves 0.0775 at the reference’s own count and 0.19 at twice it, so success budgets must bind decisions.

## 5.6 ENUMERATION-CONTRACT LEARNED RESULTS

Under the enumeration contract, the only positive learned contrast comes from one development panel, and the expanded matrix shows no gain where choice registers. On the frozen text-state BFWS development panel, one fixed-seed process-SFT run succeeds at 0.933 against 0.467 for random-valid (Table 5). The paired bootstrap 95% lower bound on that gain is 0.2, whereas random-valid saturates the BFS and additive development panels. On the expanded baseline, process SFT succeeds on 125/288 entries against 240/288 for random-valid, all in the additive cells that the identity audit reads as validity. It scores 0/24 in every BFS and BFWS cell, where choice registers, and wins none of the 72 problem-modality units in these descriptive single-seed bounds (Primary-Matrix appendices, Appendices K and L, Table 6). The BFWS positive does not reappear on this suite, and the Results Detail appendices report failure calibration, corruption, menu manipulation, image-only arms, and secondary panels outside the choice-quality claim.

## 6 DISCUSSION AND LIMITATIONS

Random-valid success separates choice from validity only after the identity audit has run on the interface at hand. The audit returns both verdicts on one runtime (Section 5.2). The ladder licenses reading a higher M1 as better choice on the validation panel and the held-out panel (Section 5.3). Be-

cause the rungs read the reference’s privileged priority order, the ladder validates the measurement, not a policy.

The adapter chooses better than uniform choice on every panel, but how far depends on the panel (Section 5.4). It is separated above the 0.75 rung only on the held-out panel. Its M1 of 0.151 to 0.432 stays well below the reference’s 0.875. On the unscreened panel the gain rests on 4 of 12 tasks (descriptive). The gain comes from imitating the exact reference.

The external audit shows that both checks are needed. On ScienceWorld (Wang et al., 2022), random-valid registers choice (Section 5.5). None of the seven surveyed published interfaces uses an enumeration contract, so the 48/48 identity failure is shown only on our runtime. On LLM-First Search, random-valid success saturates the token budget by construction although every decision diverges. Counted in expansions, the reference’s choices matter there. The contribution is therefore the measurement and its two checks, not the exposure of one flaw.

Before random-valid success is read as choice, an interface must pass two checks. The identity audit must find divergent pairs. The budget check must show that the success budget binds decisions, the compute-matched-baseline principle (Li & Talwalkar, 2020). A budget that a zero-token chooser never exhausts turns uniform choice into exhaustive randomized search. When either check fails, success reports validity or search coverage, not choice. Under a passing contract, report a policy’s difference from random-valid together with its ladder position.

### Limitations.

- *Ladder position.* The adapter is separated above the 0.75 rung only on the held-out panel, not on the validation panel, the unscreened panel, or pooled.
- *Screening.* The held-out panel is screened like the validation panel, and on the unscreened panel the gain rests on 4 of 12 tasks.
- *Seeds.* There are three training seeds, and seed 29 is weakest on every panel (descriptive).
- *Final evaluation.* The frozen 45-task final evaluation is unexecuted.
- *Prediction.* The pre-registered majority prediction for submission-order divergence failed.
- *Runtime.* The runtime never chooses or repairs operands, so no result shows unassisted internal search.
- *External audit.* The LLM-First Search reference uses a substituted local model, and the audit covers 80 tasks from four configurations.
- *Second policy.* The second realisable policy is zero-shot, and the trained second backbone was not run.

## 7 CONCLUSION

Random-valid success measures choice only on an interface that passes the identity audit and the budget check. Our runtime’s identity audit returns both verdicts, and a zero-token chooser saturates LLM-First Search’s token budget by construction. The measurement passes a pre-registered ladder test, and a trained adapter beats uniform choice across three seeds,  $D_3 = +0.285 [+0.140, +0.433]$ . On the held-out panel,  $D_3 = +0.416 [+0.250, +0.591]$ . It is separated above the 0.75 rung on the held-out panel ( $+0.227 [+0.053, +0.403]$ ) but not separated from that rung on the validation panel ( $-0.041 [-0.187, +0.102]$ ), the unscreened panel, or pooled over 35 tasks ( $+0.058 [-0.040, +0.156]$ , descriptive). The choice-quality claim is development-stage, holds relative to uniform choice and the exact- $\epsilon$  ladder on three panels with three training seeds, and carries no planning-ability or generality claim.

## REPRODUCIBILITY STATEMENT

Reproduction details are in the Trusted Search Runtime, Data and Split Procedures, Receipt Tables, and Statistics appendices. The Choice-Frontier Contract appendix specifies the choice-frontier contract. The Counterfactual, Choice-Frontier Validation, Scaled Adapter, and External Identity Audit appendices report the follow-ups. The Held-Out Panels (Appendix U.1), Dagger Ablation (Appendix U.2), and Zero-Shot Policy (Appendix U.3) appendices sit under the Scaled Adapter appendix, and the LLM-First Search Audit appendix (Appendix V.1) sits under the External Identity

Audit appendix. Table 2 holds the ledgers, and Table 3 indexes the pinned evidence paths. Code and evidence bundles will be provided as anonymous supplementary material.

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## A OPERATION SCHEMAS

This appendix gives the operation vocabulary, validity checks, and rejection kinds of each declared algorithm, for readers who need the exact interface behind the Typed Search Operation definition.

Each declared algorithm defines a vocabulary of operation types  $\tau$ . A policy emission  $u_t = (\tau_t, \theta_t)$  pairs a type from that vocabulary with an operand vector  $\theta_t$  carrying every exploration-determining argument. The runtime rejects any emission whose type falls outside the declared vocabulary or whose operand vector omits a required field. Every operand that determines exploration must originate from the policy (Section 3).

**BFS.** The primary operation is node expansion. Its operands name the source state (the current frontier head), the grounded successors selected for insertion, the action producing each successor, and the enqueue position for each. The runtime checks that the named source matches the FIFO frontier head, that each successor results from an applicable grounded action in the source state, that the successor is unvisited, and that each insertion targets the FIFO tail. A goal-test operation names the state to test. The BFS pilot corpus exposes exact grounded successor candidates and full visited membership at each step.

**BFWS (complete unpruned).** Expansion operands carry the novelty value and goal-count priority of the source state alongside the fields above. Insertion operands carry each candidate’s duplicate membership, novelty, and priority. The runtime checks that the selected source has the declared best priority, that duplicate membership is correctly reported, and that insertions follow the declared ordering. The BFWS corpus exposes candidate duplicate, novelty, priority, insertion, and enqueue facts at every training observation.

Table 1: Rejection taxonomy from the failure-mechanism calibration. Counts are failures per arm class across 2,910 replay-verified episodes (905 total failures, 0 unclassified). Share is within the arm class.

| Arm class                    | Rejection kind               | Failures | Share  |
|------------------------------|------------------------------|----------|--------|
| Base (447 ep., 447 fail.)    | malformed.output             | 379      | 84.8%  |
|                              | invalid.frontier.operation   | 67       | 15.0%  |
|                              | other.invariant.violation    | 1        | 0.2%   |
| Learned (906 ep., 407 fail.) | other.invariant.violation    | 229      | 56.3%  |
|                              | malformed.output             | 72       | 17.7%  |
|                              | inapplicable.grounded.action | 56       | 13.8%  |
|                              | effect                       | 27       | 6.6%   |
|                              | schema                       | 16       | 3.9%   |
|                              | invalid.frontier.operation   | 5        | 1.2%   |
|                              | source.state.mismatch        | 1        | 0.2%   |
|                              | static.context               | 1        | 0.2%   |
| Oracle control (51 fail.)    | expansion.budget.exhausted   | 51       | 100.0% |

**Additive best-first.** Operands carry the source state’s accumulated path cost  $g$ , heuristic estimate  $h_{\text{add}}$ , and composite value  $f = g + w \cdot h_{\text{add}}$  with  $w = 3$  for the weighted variant, or  $f = h_{\text{add}}$  for the greedy variant. Insertion operands carry per-successor  $g$ ,  $h_{\text{add}}$ , and  $f$ . The runtime checks that the expanded state has the minimum  $f$  in the frontier under the declared tie-breaking rule and that all arithmetic is consistent.

**Validity preconditions.** An emission passes through a fixed sequence of checks. Failure at any step rejects the emission with the corresponding kind and terminates the episode. The checks run in the following order.

1. **Parse.** The raw output must parse into a well-formed Typed Search Operation with a recognized type and complete operand vector. A failure is recorded as `malformed.output`.
2. **Source identity.** The named source state must match the state the algorithm expects. A failure is recorded as `source.state.mismatch`.
3. **Action applicability.** Every grounded action in the operands must be applicable in the source state. A failure is recorded as `inapplicable.grounded.action`.
4. **Successor schema.** Any named successor state must satisfy the domain schema. A failure is recorded as `schema`.
5. **Successor effects.** The named successor must match the declared effects of the applied action. A failure is recorded as `effect`.
6. **Static context.** Referenced predicates, objects, and constants must be consistent with the task instance. A failure is recorded as `static.context`.
7. **Frontier discipline.** The frontier manipulation must respect the declared ordering (FIFO for BFS, priority for BFWS,  $f$ -order for additive best-first). A failure is recorded as `invalid.frontier.operation`.
8. **Algorithm invariant.** Any remaining algorithm-specific constraint not covered above. A failure is recorded as `other.invariant.violation`.

The two broad categories are *malformed-output* rejections, where the emission never reaches the operation checker, and *post-parse rejections*, where the emission parses but fails a runtime check. The oracle-assisted random-valid control, which emits only runtime-accepted operations, has a single failure kind: `expansion.budget.exhausted`, recorded when the frozen episode budget runs out before a goal is reached.

**Rejection counts.** Table 1 gives the counts from the failure-mechanism calibration, which classified all 905 failures across 2,910 replay-verified episodes with zero unclassified. The `effect`, `schema`, and `static.context` kinds arise in successor-prediction episodes, where the model generates a dynamic-state successor.

## B ALGORITHM INVARIANT DEFINITIONS

This appendix states the invariant the runtime enforces for each declared algorithm, which Section 3 defines only in outline.

The Trusted Search Runtime checks the following invariants on every operation. Stepwise validity and invariant compliance do not establish termination, completeness, optimality, or episode success (Section 3).

**BFS.** The frontier is a FIFO queue. Every expansion must select the queue head as its source. Every insertion must target the queue tail. A state already in the visited set must not be re-inserted. This invariant is sufficient for completeness in finite state spaces with unit-cost transitions.

**BFWS (complete unpruned).** The frontier is a priority queue ordered first by novelty rank (lower is better), then by goal count among states with equal novelty. No generated successor is pruned. A successor whose state already appears in the frontier or visited set is not re-inserted.

**Additive best-first, weighted ( $w = 3$ ).** The frontier is a priority queue ordered by  $f(s) = g(s) + 3 \cdot h_{\text{add}}(s)$ , where  $g(s)$  is accumulated path cost and  $h_{\text{add}}(s)$  is the additive heuristic. Among equal- $f$  states the lower generation serial is expanded first. The expanded state is the heap head. A successor with lower  $g$  than an existing frontier entry for the same state reopens that entry with improved cost.

**Additive best-first, greedy.** The frontier is ordered by  $h_{\text{add}}(s)$  alone, with no path-cost component, then by ascending generation serial among ties. The expanded state is the heap head. Reopen and duplicate rules follow the weighted variant. In both additive settings, `BestFirstController.start_expansion` assigns serials before submissions in the deterministic `PDDLStateAuthority.applicable_actions` order (sorted by `GroundedAction.serialize`). The `BestFirstController.apply_operation` heap orders entries by (priority, serial, state identifier, path cost), and `finish_expansion` requires the full candidate set. Thus submission permutations leave frontier order unchanged on the enumeration contract. A runtime assigning serials in submission order would not be submission-order invariant and would not show this identity.

## C TRUSTED SEARCH RUNTIME

This appendix gives the runtime’s guarantees and its step and episode pseudocode, for readers checking that the runtime never chooses or repairs on the policy’s behalf.

The runtime provides three guarantees that hold across all declared algorithms.

1. **No-repair property.** The runtime validates, applies, persists to Search Memory, or rejects each emission. It never chooses omitted operands, reorders candidates, silently repairs an invalid emission, or substitutes a default search decision.
2. **Reject-and-terminate.** On the first rejected emission the episode terminates with the rejection kind recorded. There is no retry path.
3. **Replay determinism.** Replaying the raw emissions  $u_{1:T}$  against the same task instance and the same initial Search Memory reproduces the explored trace. Any step whose outcome depends on runtime repair would violate this property.

The procedures below specify the per-step check and the episode loop. No `algorithm` package is loaded. The listing uses  $\LaTeX$  tabbing.

```

procedure STEP( $u_t, M_{t-1}, A, x$ )
  // Phase 1: Parse the raw emission
  if  $u_t$  does not parse as  $(\tau, \theta)$  with  $\tau \in \text{vocab}(A)$  then
    return REJECT(malformed_output)
  // Phase 2: Validate source identity

```

```

756   if  $\theta$ .source  $\neq$  expected( $A, M_{t-1}$ ) then
757       return REJECT(source_state_mismatch)
758   // Phase 3: Validate operands against the task and state
759   for each grounded action  $a$  named in  $\theta$  do
760       if  $a$  is not applicable in  $\theta$ .source under  $x.D$  then
761           return REJECT(inapplicable_grounded_action)
762   for each successor state  $s'$  named in  $\theta$  do
763       if  $s'$  violates the domain schema then
764           return REJECT(schema)
765       if  $s' \neq$  effects( $a, \theta$ .source) then
766           return REJECT(effect)
767   if any static-context reference in  $\theta$  is inconsistent with  $x$  then
768       return REJECT(static_context)
769   // Phase 4: Check Algorithm Invariant
770   if frontier discipline of  $A$  is violated by  $(\tau, \theta, M_{t-1})$  then
771       return REJECT(invalid_frontier_operation)
772   if any other invariant of  $A$  is violated then
773       return REJECT(other_invariant_violation)
774   // Phase 5: Apply and persist
775    $M_t \leftarrow$  apply( $\tau, \theta, M_{t-1}$ )
776   return ACCEPT( $M_t$ )
777
778 procedure EPISODE( $x, A, \phi_m, B$ )
779    $M_0 \leftarrow$  InitSearchMemory( $A, x$ )
780   for  $t = 1, 2, \dots, B$  do
781        $o_t \leftarrow \phi_m(c_t(x, M_{t-1}))$  // build observation
782        $u_t \leftarrow \pi(o_t)$  // policy emission
783        $r \leftarrow$  STEP( $u_t, M_{t-1}, A, x$ )
784       if  $r$  is REJECT(kind) then
785           return ( $t, \text{FAIL}, \text{kind}$ )
786        $M_t \leftarrow r.M$ 
787       if GoalTest( $M_t, x.G$ ) = TRUE then
788           return ( $t, \text{SUCCESS}$ )
789   return ( $B, \text{FAIL}, \text{expansion\_budget\_exhausted}$ )

```

Each call to STEP consumes one unit of the frozen episode budget  $B$ , whether the emission is accepted or rejected. Invalid operations are therefore charged to the budget (Section 4).

## D DATA AND SPLIT PROCEDURES

This appendix describes the corpora, splits, and evaluation manifests behind the reported panels, for readers checking that no evaluation task leaked into training.

All training and evaluation splits are by whole task instance. No state, decision, or target from an evaluation task appears in any training row. Every released corpus passed no-overlap, no-conflict, live/training-parity, and token-budget audits before use.

**BFS pilot corpus.** The v6 text-state process corpus holds 25,109 process rows split into 12,994 train and 12,115 dev rows, drawn from 90 replay-verified trusted traces covering 90 distinct tasks. The release assigns splits by semantic task identity, with zero cross-split task overlap. The v8 evaluation panel selects 15 tasks from this corpus’s development set, not its training set. No v8 evaluation-panel task contributes training rows, consistent with the whole-instance rule above. Corpus observations expose the canonical task context, exact grounded successor candidates, and full visited membership. Release audits recorded zero semantic, input, and input-target overlap, zero conflicting identical-input labels, zero teacher-operation rejections, and zero live/corpus mismatches, all within an 8,192-token input cap and a 384-token output allowance.

**BFWS corpus.** The corpus covers the frozen 105-task development panel with 69,019 replay-verified teacher decisions. Its release contains 69,019 process records, 67,215 operational records, and a process projection of 47,780 train and 21,239 dev rows. Each step’s observation exposes

Table 2: Combined per-branch gate receipts and ledger allocations. The expanded program spent 68.52 of 336 GPU-h, rounded independently by row. The second-backbone cap reflects a prospective 12.11 GPU-h transfer from recovery reserve. Corruption and seed replication draw on existing branch caps. Follow-up windows have separate ledgers and are never summed into the program total or with each other. The R1-R4 spend covers both stress-test rows.

| Branch                                                                 | Tickets         | Terminal state                             | Frozen scope / replay coverage                                                                                             | GPU-h / cap           |
|------------------------------------------------------------------------|-----------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Expanded baseline                                                      | #116–#118       | complete                                   | 1,152 bindings, 576 model, 1,152/1,152 replayed                                                                            | 6.53 / 56             |
| DAGger                                                                 | #78–#84         | complete                                   | 405 episodes + 81 paired rows, 405/405                                                                                     | 10.91 / 48            |
| Successor prediction                                                   | #85–#89, #99    | complete                                   | 90 episodes, 15 cells, 90/90                                                                                               | 10.85 / 64            |
| Curriculum by modality                                                 | #119            | complete                                   | 243 model + 324 comparator bindings, 567/567                                                                               | 11.78 / 48            |
| Transfer                                                               | #104–#107       | complete                                   | 39 cells (12 v1, 27 v2), 39/39                                                                                             | 3.60 / 24             |
| Second backbone                                                        | #101–#103, #123 | v1 VALID_STOP, v2 complete                 | 72 model + 72 comparators, 144/144                                                                                         | 6.03 / 52.11          |
| Generalization / robustness                                            | #96, #98, #124  | v1 VALID_STOP, v2 complete                 | 1,200 episodes (300 model, 900 controls), 1,200/1,200                                                                      | 6.04 / 32             |
| Observation corruption                                                 | #126            | complete                                   | 360/360 new bindings replayed                                                                                              | within generalization |
| Training-seed replication                                              | #127            | complete                                   | 534/534 new episodes                                                                                                       | 12.79 / –             |
| Recovery reserve                                                       | –               | reduced                                    | –                                                                                                                          | 0.00 / 11.89          |
| <b>Total</b>                                                           |                 |                                            |                                                                                                                            | <b>68.52 / 336</b>    |
| <i>Follow-up windows (separate ledgers, not summed with the total)</i> |                 |                                            |                                                                                                                            |                       |
| Native arms                                                            | #128, #129      | v1 gate history, v2 complete               | 216/216 new + 72/72 comparators                                                                                            | 3.2086 / 12           |
| Stress tests R1/R2/R4                                                  | #130            | complete                                   | 72 menu + 80 variant + 18 cross-cell, 170/170                                                                              | 7.7998 / 36           |
| Second backbone v3 (R3)                                                | #130            | complete                                   | 72/72 model + 72/72 comparators                                                                                            | same window           |
| Choice frontier                                                        | #132            | CHOICE_SENSITIVE                           | 144/144 panel, 58 preparation episodes re-derived                                                                          | 1.069 / 12            |
| Comparator zoo                                                         | #133            | complete                                   | 432/432 bindings, 342 new                                                                                                  | CPU only              |
| Counterfactual audit, M4 intervals                                     | #134            | prediction failed                          | 48 pairs, 288 counts + 96 sequences re-validated                                                                           | CPU only              |
| Instrument validation                                                  | #135            | PASS                                       | 696/696 CPU + 48/48 model episodes replayed                                                                                | 1.19 / 4              |
| Scaled adapter                                                         | #136            | POSITIVE                                   | 24/24 learned episodes replayed                                                                                            | 4.84 / 30             |
| External audit                                                         | #137            | CHOICE_REGISTERED                          | 1000 pairs over 200 tasks                                                                                                  | CPU only              |
| Three-seed replication                                                 | #138, #139      | POSITIVE                                   | 48/48 new learned episodes, 638/638 P2 and 696/696 P2u zoo episodes replayed, and 88/88 P2 plus 96/96 P2u learned episodes | 15.22 / 24            |
| held-out panels (v4 window)                                            |                 |                                            | 24/24 validation-panel and 22/22 P2 episodes replayed                                                                      | 9.06 / 40             |
| DAGger ablation (v5 window)                                            | #140            | NOT_SEPARATED                              | 46/46 model episodes replayed                                                                                              | 2.25 / 30             |
| Zero-shot second policy (v6 window)                                    | #141            | EQUIVALENT (validation), INCONCLUSIVE (P2) | 80/80 reference and 400/400 random-valid episodes replayed                                                                 | 2.89 / 12             |
| LLM-First Search audit (external v2)                                   | #142            | SATURATED (paper budget)                   |                                                                                                                            |                       |

candidate duplicate membership, novelty values, goal-count priorities, insertion positions, and enqueue facts. Audit counters recorded zero overlap, mismatch, rejection, or final-evaluation-access violations.

**Evaluation manifests.** The BFS pilot uses a frozen 15-task outcome-blind development panel. The BFWS structural gate froze a 105-task development panel and, separately, the manifest of the frozen 45-task final evaluation. That manifest has never been accessed, and no model prediction, training decision, or analysis references it. The expanded baseline evaluates 24 tasks from 12 domains under 4 algorithms, 3 modalities, and 4 arms (1,152 logical bindings, 576 model episodes, all independently replayed). The curriculum stage evaluates 27 unseen tasks. The generalization-v2 protocol admits 25 derived tasks from 93 eligible semantics-preserving variants (uniform  $k=5$  cheapest-prefix per family, membership frozen before execution).

## E RECEIPT TABLES

This appendix holds the gate receipts, the per-window ledgers, and the artifact index, so that a reader can trace each reported run to its receipt.

Table 2 combines branch-level frozen scope, gate status, coverage, and compute from the synthesis and follow-up ledgers. Five branches completed at v1 scope. Two stopped at frozen v1 cost-admission gates with terminal VALID\_STOP receipts and later completed at reduced v2 scope. Every branch reconciles with zero missing evidence. The Tickets column indexes internal ticket numbers.

At the synthesis snapshot, the program recorded 135 attempts (96 succeeded, 36 failed, 3 cutoff). Failed and cutoff attempts retain their hours in the ledger. Every attempt ended before the absolute scheduler cutoff of 2026-09-21T11:55:19Z.

The observation-corruption suite added 2.22 GPU-h to the generalization branch. The training-seed replication addendum (closed 2026-09-21) added 12.79 GPU-h and 534 independently replayed episodes covering two cells. Together these addenda brought the program total to 68.52/336 GPU-h. The Statistics appendix reports the seed-variance results.

**Follow-up windows.** Each follow-up window keeps its own ledger, and none is added to the program’s 68.52/336 GPU-h. The native-arms window (v1 probes, then v2 training and evaluation) spent 3.2086 of 12 GPU-h. Its 216 new bindings (144 model episodes and 72 CPU random-valid episodes) and 72 reused comparators were all independently replayed. The R1-R4 follow-up window spent 7.7998 of 36 GPU-h across two ledgers. It was resized from 20 GPU-h before any outcome, because its frozen worst-case estimate of 27.04 GPU-h including spend exceeded the initial cap, and realized cost landed at 29% of that estimate. The choice-frontier window spent 1.069 of 12 GPU-h (training 0.81, smoke 0.052, model evaluation 0.208), with corpus derivation and controls on CPU. Its 144 panel episodes were replayed and its 58 preparation episodes re-derived bit-exact. The comparator zoo launched no GPU job. It replayed all 432 zoo bindings, of which 342 are new CPU episodes and 90 are reused  $2\times$  random-valid controls. The three-seed replication and the held-out panels shared one window, which spent 15.22 of 24 GPU-h, and the DAgger ablation window spent 9.06 of 40 GPU-h, including 1.79 GPU-h charged for a cancelled collection. The zero-shot window spent 2.25 of 30 GPU-h, and the LLM-First Search audit spent 2.89 of 12 GPU-h on its reference server with no paid API use. Failed attempts retain their hours in every window.

**Artifact index.** Table 3 maps each pinned follow-up evidence path to the class of claim it supports. Paths are relative to the evidence repository.

## F STATISTICS

This appendix states the interval, test, margin, and tiny-strata conventions behind every reported contrast, for readers judging what an interval licenses.

**Default contrast procedure.** Contrasts are paired by whole task instance. Paired differences use a bootstrap with 10,000 resamples at seed 1729 and 95% percentile intervals. Under one training seed and panels of 24 or fewer instances, these intervals are descriptive bounds. They do not support broad superiority claims.

The BFWs gate used confidence level 0.95 with 10,000 paired whole-instance resamples (bootstrap seed 1729). Its reported lower bound is the 0.025 quantile, i.e., the lower endpoint of a two-sided 95% percentile interval.

**Tiny-strata rule.** Strata with fewer than 8 episodes are reported as descriptive only. No intervals are computed over such strata. They are listed explicitly and never silently pooled or dropped.

**Transfer tests.** Transfer comparisons use McNemar exact tests against the per-example base predictions, with Holm correction across all 36 adapter-by-benchmark comparisons. The minimum adjusted  $p$  in the executed transfer branch is 0.8156. None survives correction.

**Follow-up bootstrap conventions.** The follow-up windows fix their own bootstrap seeds, separate from the main program’s seed 1729 default. The native-arms contrasts and the choice-frontier contrasts use a paired whole-problem bootstrap at seed 61813 with 10,000 resamples and 95% percentile intervals. The second-backbone v3 contrasts (stress test R3) use seed 1729 with 10,000 resamples. The comparator-zoo metrics use a task-cluster bootstrap that draws nine tasks with replacement at seed 133 for 10,000 draws. Both additive cells and all random seeds stay inside their task cluster.

**Materiality and non-equivalence.** A follow-up contrast is material when its 95% interval excludes zero. The follow-up protocols through the comparator zoo pre-registered no equivalence margin, so a non-material contrast there supports neither superiority nor equivalence. The later choice-frontier protocols pre-register an equivalence margin of  $\pm 0.05$  for  $D$  and for the DAgger change  $\Delta$ , and none for the separation  $S$ , so a non-separated  $S$  is not equivalence to the rung. Wherever the text reports a non-material contrast or a comparison that fails Holm correction, an adjacent

Table 3: Artifact index for the follow-up evidence. Each row names a pinned path and the claim class that rests on it.

| Pinned path                                                                                                                                                                                                                      | Claim class                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| outputs/native-arms/v1/evaluation/<br>(evaluation.json, analysis.json)                                                                                                                                                           | Native-arms clean and corruption success,<br>paired contrasts                             |
| outputs/native-arms/v1/identity-audit.<br>json                                                                                                                                                                                   | Identity finding on the enumeration contract                                              |
| outputs/native-arms/v1/<br>stress-evaluation.json                                                                                                                                                                                | Menu manipulation, degraded strains, R2<br>cross cell, stress replay                      |
| outputs/native-arms/v1/r2-decomposition.<br>json                                                                                                                                                                                 | Text-corruption termination modes                                                         |
| outputs/native-arms/v1/budget.json and the<br>R1-R4 window ledgers (v1, v2)                                                                                                                                                      | Native-arms and R1-R4 window ledgers                                                      |
| outputs/expanded-study/v1/<br>second-backbone-v2/evaluation/<br>outputs/expanded-study/v1/<br>second-backbone-v3/evaluation/<br>outputs/choice-frontier/v1/evaluation/<br>(evaluation, identity-audit, analysis,<br>final-audit) | Second-backbone BFS cell<br><br>Second-backbone additive replication and its<br>contrasts |
| outputs/choice-frontier/v1/preparation/<br>visual-choice-frontier/ (report, audit)                                                                                                                                               | Choice-frontier gate, panel rows, contrasts, re-<br>play                                  |
| outputs/choice-frontier/v1/budget.json                                                                                                                                                                                           | Choice-frontier corpus derivation                                                         |
| outputs/choice-frontier/o4/zoo/manifest.<br>json                                                                                                                                                                                 | Choice-frontier window ledger                                                             |
| outputs/choice-frontier/<br>o4/metrics/ (analysis,<br>stored-posthoc-analysis, optimal-costs,<br>native-four-algorithm-reference-costs)                                                                                          | Zoo coverage and replay                                                                   |
| docs/experiments/native-arms/<br>issue-129-closeout.md,<br>issue-130-closeout.md                                                                                                                                                 | Exploratory M1–M5 metrics, Holm family,<br>optimal costs, native reference costs          |
| docs/experiments/choice-frontier/<br>issue-132-closeout.md                                                                                                                                                                       | Native-arms and stress-test R1-R4 records                                                 |
| docs/experiments/choice-frontier/<br>metric-adjudication-2026-09-23.md                                                                                                                                                           | Choice-frontier outcome and its binding erra-<br>tum                                      |
| outputs/choice-frontier/v4/seeds/<br>metrics/analysis.json                                                                                                                                                                       | Post-close metric adjudication                                                            |
| outputs/choice-frontier/v4/panels/<br>metrics/analysis.json                                                                                                                                                                      | Three-seed endpoint, separation, seed variance<br>on the validation panel                 |
| outputs/choice-frontier/v5/metrics/<br>analysis.json                                                                                                                                                                             | Held-out and unscreened panels, per-panel<br>arms, pooled results, concentration          |
| outputs/choice-frontier/v4/budget.json,<br>outputs/choice-frontier/v5/budget.json                                                                                                                                                | DAGger ablation                                                                           |
| outputs/choice-frontier/v6/metrics/<br>analysis.json                                                                                                                                                                             | v4 and v5 window ledgers                                                                  |
| outputs/choice-frontier/v6/budget.json                                                                                                                                                                                           | Zero-shot second policy                                                                   |
| outputs/external-audit/v2/lfs/<br>identity-audit.json                                                                                                                                                                            | v6 window ledger                                                                          |
| outputs/external-audit/v2/budget.json                                                                                                                                                                                            | LLM-First Search audit                                                                    |
| Comparator-zoo ticket (#133) closing comment                                                                                                                                                                                     | External audit v2 ledger                                                                  |
|                                                                                                                                                                                                                                  | Zoo numbers table and claim boundary                                                      |

sentence states that non-significance is not equivalence. The exact sign-flip  $p$  value enumerates all  $2^n$  sign vectors of the  $n$  per-task differences, is two-sided, tests a zero difference, and is descriptive, so it never supports equivalence.

**Tiny strata in the follow-ups.** The fewer-than-8 rule also governs discriminative subsets. The choice-frontier learned-minus-random-valid contrast is computed over 18 cells, but it rests entirely on storage, the only panel task where random-valid’s seed frequency lies strictly between 0 and 1 (0.2). A one-task discriminative subset falls under the rule, so that contrast is descriptive. The zoo’s M5 subset, defined from controls only, holds the same single task and is reported descriptively.

Table 4: Seed-variance replication for Cell A (the greedy  $\times$  multimodal replication cell). Contrasts are paired whole-instance bootstrap (24 tasks, 10,000 resamples, seed 1729). The primary contrast (SFT – base) is positive at every seed with no sign flip.

| Seed | Successes / 24 | SFT – base | SFT – random-valid |
|------|----------------|------------|--------------------|
| 17   | 21/24          | +0.875     | −0.125             |
| 29   | 21/24          | +0.875     | −0.125             |
| 71   | 22/24          | +0.917     | −0.083             |

**Zoo dominance tests.** Any positive learned-versus-comparator dominance claim in the comparator zoo uses paired task-cluster bootstrap one-sided  $p$  values with Holm step-down correction over five comparators and three metrics, fifteen tests in all. Undefined or uninformative co-solved contrasts prohibit a positive dominance claim and never count as wins. All zoo metrics are exploratory, and those computed on the stored choice-frontier episodes are post-hoc.

**Seed-variance addendum.** A prospective training-seed replication measured variance for exactly two cells. Seeds 17, 29, and 71 were frozen before any seed-29 or seed-71 artifact existed. All 534 new episodes were independently replayed. Cross-seed aggregation is descriptive ( $n = 3$  seeds).

Cell A is the greedy  $\times$  multimodal replication cell (`best_first_add_greedy`  $\times$  multimodal-state) evaluated on the 24-problem panel. Each seed trained a fresh LoRA adapter with new initialization and dropout randomness on the identical frozen 512-record membership. Table 4 gives the per-seed results.

Cell set B covers DAGger iteration-1 versus the exposure-matched continued-SFT control under BFS across all three modalities. The null replicates, with 0/72 unseen successes for every arm and every seed. The paired contrast (DAGger minus continued SFT) is +0.000 [0.000, 0.000] at each seed.

This addendum supersedes the blanket single-training-seed boundary for these two cells only. Every other enumeration-contract cell in the program remains single-seed (seed 17) evidence.

## G MODALITY ADAPTERS AND OBSERVATION CONTRACT

This appendix defines the observation adapters and what each one exposes to the policy, which bounds any reading of the modality results.

Three fixed Modality Observation adapters are declared. Each maps a decision context  $c_t$  (the canonical task context plus the decision-relevant view of Search Memory  $M_t$ ) to an observation  $o_t = \phi_m(c_t)$  presented to the policy. The adapter is fixed for an entire training or evaluation cell. The adapter contract requires decision sufficiency, meaning that  $o_t$  must expose every fact the exact reference uses to resolve step  $t$  under the declared algorithm.

**text-state.** The state, goal, and search context are serialized as a compact relational representation. Object identities, predicate names, and all Search Memory contents (frontier, visited set, algorithm-specific state) appear as structured text.

**visual-state.** The state and goal are rendered as a pair of 128 $\times$ 128 pixel images, a rendered state image and a partial-goal constraint image. Object labels are not embedded in the rendered image. Search Memory contents and the grounded candidate list remain in the textual serialization, even in this adapter. This is the published frozen `visual-state` arm that the native arms below are compared against.

**multimodal-state.** The observation combines the text-state serialization and the visual-state rendered images. The policy receives the relational text and the image pair jointly.

**Native image-only arms.** Two further adapters executed under the native-arms v2 contract. `visual-nomem-state` keeps only the static task interface and the grounded, unscored candidate menu as text, and carries state facts in the current rendered scene alone. `visual-seq-state` adds up to  $K=8$  rendered history frames, each labelled with its absolute path depth. Neither receives textual Search Memory, accepted deltas,  $g/h$ /priority scalars, candidate scores, or membership flags.

**Text-scaffold caveat.** Search Memory contents and the candidate list are serialized as structured text in all three declared adapters, including visual-state. Rendered images carry the planning state and goal visuals, but search bookkeeping (frontier, visited set, novelty counters, candidate list) always travels through the text channel. The observation-corruption suite is consistent with this. Copyability-preserving text shuffling drops success by up to  $-0.944$  (text) and  $-0.722$  (multimodal), while visual corruption costs at most  $-0.111$  with intervals that include zero.

The native arms narrow this caveat to the candidate menu. With the textual bookkeeping and scalars removed, both arms solve 18/18 clean tasks with zero invalid operations, and blanking or degrading every image of the seq arm gives a corrupted-minus-clean contrast of  $0.000$   $[0.000, 0.000]$ . As a post-hoc interpretation, the grounded unscored menu is the remaining text channel that could carry the signal. Menu manipulation shows that the policies do not reject schema-valid inapplicable menu entries (pick-rate at chance), and on the additive cells submission-order invariance makes any listed candidate a correct emission. The caveat therefore still holds for every adapter that presents the menu, and no executed adapter isolates an image-only decision channel.

**Information-loss boundaries.** The 128-pixel unlabelled rendered views can lose symbolic identities. The P3 name-compression family in the generalization-v2 suite is information-lossy. Family-wide (24 eligible P3 variants), 24/24 variants are text-lossy, 18/24 are multimodal-lossy, and 0/24 are visual-lossy. Modality matching between training and evaluation ensures identical observation format, not information parity. Static-context and goal fields of the observation contain text in every adapter. The available evidence does not establish semantic parity across adapters or an intrinsic modality effect.

## H RETAINED-INFRASTRUCTURE FEASIBILITY NOTES

The terms below describe verified infrastructure components that support corpus generation, evaluation, and evidence production. They constitute feasibility evidence for the runtime and rendering pipeline. They cannot support a Search Process Policy efficacy claim.

**Plan pipeline.** **Plan Submission** transmits a supplied plan from the project adapter to a planning or rendering backend. **Plan Interpretation** converts the submitted plan into the backend’s ordered action representation. An **Action Sequence** is the canonical identity of a supplied plan as a normalized, ordered sequence of grounded action predicates. Casing and whitespace are not part of its identity.

**Rendering pipeline.** **Render Production** creates a visualisation output and its rendered image for a planning state. **Render Validation** confirms that the produced visualisation and image satisfy declared structural and semantic checks. **Plan Provenance** links a produced visualisation to the exact supplied plan that the backend interpreted.

**Certification and evidence.** **Integration Certification** is the finding that Plan Submission, Plan Interpretation, Plan Provenance, Render Production, Render Validation, and required execution-isolation checks all passed. An **Attempt** is one authorized, immutable execution with its own identity and retained outcome, whether successful or not. An **Evidence Bundle** is the self-contained records from an Attempt that allow an independent reproducer to verify its claims offline. The retained infrastructure supplies deterministic compact delta/event evidence with semantic replay while preserving search semantics, serving the replay-determinism requirement of the Trusted Search Runtime.

**Historical terms.** Planning Certificate, Joint Action-and-Certificate SFT, Adaptive Scaffolding, Support Route, Live Memory, Verified Joint Step, and Route Label are historical CGAS terms. They describe retained infrastructure internals. They cannot support any current Search Process Policy efficacy claim and are not referenced elsewhere in this manuscript.

## I CHOICE-FRONTIER CONTRACT AND COMPARATOR SELECTORS

This appendix specifies the choice-frontier contract, its corpus derivation, its metrics, and the comparator selectors, for readers reproducing the measurement that Section 5.3 validates.

**Menu and observation.** At each decision the runtime exposes the live frontier as an unscored menu of opaque labels  $c_0, \dots, c_K$ . Labels are assigned after a seeded Fisher–Yates permutation of the generation-serial member order, with a per-decision seed derived from master seed 51131 and a hash of the member state identifiers, so a label carries no positional information. The observation holds static task-context pages, the unlabelled 128px initial-state scene, one unlabelled 128px scene per frontier state in menu order, and the partial-goal pages. Its text payload carries only the algorithm name, the label list, the representation tag, the schema version, and the frozen legend. A leak check fails closed on any forbidden key marker.

**Output and validation.** The policy emits canonical JSON with the single key `expand_choice` and a label value. The runtime rejects malformed JSON, a wrong key set, a non-string label, a label not offered by the pending menu, and a label bound to a non-live member. The first rejection ends the episode as `deterministic_invalid_operation`. Other terminations are `goal_reached`, `frontier_exhausted`, `decision_budget_exhausted`, and `expansion_budget_exhausted`.

**Expansion and budget.** On acceptance the runtime removes the chosen member from the frontier and expands it atomically. Candidate generation, serial assignment, admission, duplicate detection, reopening, and expansion bookkeeping are inherited unchanged from the frozen additive best-first controller, which is not modified. Per-candidate admissions are runtime-internal, so one submitted choice is one decision. Selecting a member that satisfies the goal solves the task without an expansion and without budget charge, so the exact episode takes its reference expansion count plus one decisions. Decision and expansion caps both equal  $2 \times$  the reference expansion count.

**Corpus derivation.** The training corpus was derived on CPU by replaying the frozen exact traces through the new contract and verified expansion-for-expansion against them (58 episodes bit-exact). The frozen matched-modalities membership indexes enumeration-contract decisions. Each such decision maps to the expansion whose candidate set it submitted into, unique (task, expansion) pairs in first-occurrence order form the base set, and a round-robin over the frozen task order tops the set up to 512 training records per algorithm. Diagnostics are the unique expansions of the frozen diagnostic identifiers, with no top-up. The teacher target is the label bound to the heap head. Five fail-closed gates run before training. They check expansion-for-expansion agreement with the stored exact traces, a token budget of measured input plus 384 within 32,768, the text-leak check, image resolution, and identity with the frozen membership. The pinned preparation report records 29 tasks, 1,046 records (512 training and 11 diagnostic per algorithm), 3,544 scenes, and 58 episodes re-derived bit-exact, with outcome PASS. The smoke gate accepted 40/40 model calls.

**Training and replay.** Each of the two cells trains a LoRA adapter (rank 64,  $\alpha$  128) for 16 updates on its 512 records at seed 17 with the frozen expanded-program recipe. Both training jobs completed all 16 steps and saved final adapters, then failed in post-save summary construction. Their reports were recovered from the primary training-state artifacts with provenance marked, no retraining was needed, and the training audit passes on the recovered reports. At finalize, independent replay re-derives every menu binding, payload, and trusted transition from the stored events, and checks pixel bindings against the retained scenes.

**Comparator selectors.** Each selector’s choice is mapped back to the menu binding, and every candidate still enters through the additive admission path. Frontier admission, duplicate detection,  $h_{\text{add}}$  computation, reopening, and serial ordering therefore stay additive for `bfs_order` and novelty-first selection alike. The novelty features of novelty-first selection are positive PDDL state atoms, and its seen-feature sets advance only when a state is expanded, including the initial expansion. Each new episode is re-run against an independently constructed PDDL authority, which checks menus, decisions, admissions, and result dictionaries. Selector episodes replay the model input, menu, actions, admissions, and budget, but they neither generate nor replay image bindings. The optimal

costs  $C^*$  come from one offline shortest-plan computation per task with replay-verified plans, and the native four-algorithm reference costs are recorded separately from the zoo.

**Metric computation.** Under M1 an episode solves at multiplier  $m$  when its first goal selection occurs within  $\lfloor mR_t \rfloor$  decisions with at most  $\lfloor mR_t \rfloor$  preceding expansions, where  $R_t$  is the additive exact-reference expansion count. Unsolved  $2\times$  runs count as failures at every  $m$ , and the same prefix truncation applies to every arm. Task and algorithm weights are equal. M2 also has a censored form, in which failures are right-censored at the budget rather than assigned a fictitious expansion count. M3 counts transitions along the selected goal state’s trusted predecessor chain and is undefined, never imputed, for unsolved episodes. M4 measures agreement with the current additive heap head over decisions with at least two live choices, against a chance rate equal to the decision-weighted mean of  $1/K$ . M5 restricts to tasks where the mean  $2\times$  random-valid solve indicator lies strictly between 0 and 1. Because goal selection is the  $(R_t+1)$ -th decision, the exact reference fails at  $m = 1$  and scores  $M1 = 0.875$  by construction, and arms that reach the goal in fewer decisions can exceed it (privileged hadd-greedy 0.882, 0.928, and 0.906).

## J RESULTS DETAIL: GATE RECEIPTS AND QUALIFICATION GATES

This appendix gives the program’s receipts, the follow-up ledgers, and the qualification gates that Section 5.1 summarizes.

The expanded study program ran on 2 A100 80GB GPUs under a 336 GPU-hour cap with an absolute scheduler cutoff, and spent 53.52 of those hours (15.9%) at the synthesis snapshot, before the corruption and seed-replication addenda, and 68.5244 after them. Seven branches were admitted. Five completed at v1 scope, and two (second backbone, generalization/robustness) stopped at frozen v1 cost-admission gates with terminal VALID\_STOP receipts, then completed at reduced v2 scope under new versioned protocols. Table 2 combines frozen scope, status, coverage, and ledger spend. Every executed branch reconciles with zero missing evidence, with 1,152/1,152 baseline episodes independently replayed, 405/405 DAGger episodes, and 90/90 successor-prediction episodes.

Follow-up windows executed after the program on separate GPU ledgers (Table 2) and on CPU only (the comparator zoo, the counterfactual audit, and the first external audit). These windows are reported separately and never summed with the main program ledger. The native-arms window spent 3.2086 of 12 GPU-hours, the R1-R4 follow-up window spent 7.7998 of 36, and the choice-frontier window spent 1.069 of 12. The R1-R4 follow-up window was resized from 20 to 36 GPU-hours before any outcome, because its frozen worst-case estimate of 27.04 GPU-hours including spend exceeded the initial cap, and realized cost landed at 29% of that estimate. Each follow-up closed with complete coverage and independent replay (Table 2).

Three earlier gates frame the active algorithm matrix. The bounded BFS pilot closed with a governed VALID\_STOP. On a frozen outcome-blind 15-task BFS development panel, final checkpoints from five process-SFT seeds achieved 1.0 invariant-valid success with a 0.0 invalid-operation rate, but the base achieved 0.0 and random-valid achieved 1.0, so the learned policy showed executable BFS behavior with zero gain over the strongest control and downstream work was gated off. The BFWS structural gate then passed on its frozen development panel (Section 5.6). The companion manifest of the frozen 45-task final evaluation was fixed before outcomes and has never been accessed. The optimal A\* arm over  $h_{\max}$  with landmark counting was retired after its own governed VALID\_STOP and is superseded by the two additive best-first variants. Its original heuristic-representation estimand remains unidentified. The historical five-seed BFS pilot predates the budget rule, which authorizes one training run at seed 17 per enumeration-contract model and cell (Section 4).

## K RESULTS DETAIL: PRIMARY-MATRIX TABLES

Table 5 gives the BFWS development panel and Table 6 the expanded 24-task baseline by algorithm and modality, both summarized in Section 5.6. Measures are reported separately and never combined.

Table 5: Frozen 15-task text-state BFWS development panel, with full-episode budgeted success by condition. The learned-to-best-control gain over random-valid is 0.467 with a paired whole-instance bootstrap 95% interval lower bound 0.2 (10,000 resamples). Single fixed training seed, development stage, final-evaluation manifest not accessed.

| Condition                              | Success (15 tasks) |
|----------------------------------------|--------------------|
| Pretrained base                        | 0.0                |
| Process SFT                            | 0.933              |
| Random-valid (oracle-assisted control) | 0.467              |
| Exact execution                        | 1.0                |

Table 6: Primary matrix of the expanded 24-task baseline by algorithm and modality. Measures are reported separately and never combined. Columns give full-episode budgeted success out of 24 for each condition, process-SFT invalid operations over decisions (inv/dec), process-SFT expansions, and exact-reference decisions as the budget reference. The pretrained base terminates on its first invalid operation in every cell (24/24 inv/dec, 0 expansions). Random-valid (RV) is an oracle-assisted programmatic control and exact execution is the oracle upper bound, and both are references rather than model abilities.

| Algorithm             | Modality   | Base | SFT   | RV    | Exact | SFT inv/dec | SFT exp. | Exact dec. |
|-----------------------|------------|------|-------|-------|-------|-------------|----------|------------|
| BFS                   | text       | 0/24 | 0/24  | 15/24 | 24/24 | 24/66       | 15       | 1969       |
| BFS                   | visual     | 0/24 | 0/24  | 15/24 | 24/24 | 24/69       | 17       | 1969       |
| BFS                   | multimodal | 0/24 | 0/24  | 15/24 | 24/24 | 24/63       | 15       | 1969       |
| BFWS                  | text       | 0/24 | 0/24  | 17/24 | 24/24 | 24/24       | 0        | 974        |
| BFWS                  | visual     | 0/24 | 0/24  | 17/24 | 24/24 | 24/24       | 0        | 974        |
| BFWS                  | multimodal | 0/24 | 0/24  | 17/24 | 24/24 | 24/24       | 0        | 974        |
| Weighted A* ( $w=3$ ) | text       | 0/24 | 20/24 | 24/24 | 24/24 | 4/609       | 179      | 694        |
| Weighted A* ( $w=3$ ) | visual     | 0/24 | 18/24 | 24/24 | 24/24 | 6/499       | 143      | 694        |
| Weighted A* ( $w=3$ ) | multimodal | 0/24 | 22/24 | 24/24 | 24/24 | 2/666       | 196      | 694        |
| Greedy best-first     | text       | 0/24 | 22/24 | 24/24 | 24/24 | 2/614       | 179      | 673        |
| Greedy best-first     | visual     | 0/24 | 22/24 | 24/24 | 24/24 | 2/643       | 189      | 673        |
| Greedy best-first     | multimodal | 0/24 | 21/24 | 24/24 | 24/24 | 3/617       | 179      | 673        |

## L RESULTS DETAIL: PRIMARY-MATRIX PAIRED CONTRASTS

Paired whole-problem contrasts supply the learned-to-best-control gain, kept separate from the success counts (Table 7). Against the base, process SFT gains only in the additive cells (+0.833 for weighted A\*, +0.903 for greedy) and gains nothing under BFS or BFWS. Against random-valid, process SFT loses in every algorithm column and wins zero of the 72 problem-modality units in any column. Random-valid itself ties exact execution in both additive columns, so those cells cannot measure any advantage beyond valid bookkeeping. These contrasts are tiny-subgroup descriptive bounds over 24 problems and one training seed. They do not support modality or algorithm superiority claims beyond the executed panel. The development-panel BFWS positive does not reappear on this suite. The complete six-measure matrix for the frozen final protocol, including per-cell Algorithm Invariant compliance and charged-budget columns, belongs to the unexecuted final evaluation.

## M RESULTS DETAIL: SECONDARY PANELS

This appendix reports the secondary enumeration-contract panels, which Section 5.6 points to and which sit outside the choice-quality claim.

**Modality.** The per-modality rows of Table 6 show the same qualitative profile in all three adapters, with zero learned success under BFS and BFWS and learned success below the saturating controls in the additive cells. The earlier matched-modality study, a 3-problem panel on its own ledger that is never pooled with the expanded baseline, gives process-SFT success of 5/12 in text-state, 6/12 in visual-state, and 6/12 in multimodal-state, versus 10/12 for random-valid in each modality, 0/12 for the base, and 12/12 for exact execution. The evidence does not support an intrinsic modality effect. Two caveats bound any such claim. The unlabelled 128px rendered views can lose symbolic identities, and the static and goal pages still contain text in every modality, so semantic parity across adapters is not established.

Table 7: Paired whole-problem success contrasts on the expanded baseline (10,000 paired resamples over the 24 problems, seed 1729). Entries give the point contrast with wins/losses/ties over the 72 problem-modality units. Contrasts are descriptive bounds over a tiny subgroup and a single training seed (Efron, 1979).

| Contrast             | BFS              | BFWS             | Weighted A* ( $w=3$ ) | Greedy          |
|----------------------|------------------|------------------|-----------------------|-----------------|
| SFT – base           | +0.000 (0/0/72)  | +0.000 (0/0/72)  | +0.833 (60/0/12)      | +0.903 (65/0/7) |
| SFT – random-valid   | −0.625 (0/45/27) | −0.708 (0/51/21) | −0.167 (0/12/60)      | −0.097 (0/7/65) |
| SFT – exact          | −1.000 (0/72/0)  | −1.000 (0/72/0)  | −0.167 (0/12/60)      | −0.097 (0/7/65) |
| Random-valid – exact | −0.375 (0/27/45) | −0.292 (0/21/51) | +0.000 (0/0/72)       | +0.000 (0/0/72) |

Table 8: DAgger on BFS, with development and unseen success and unseen invalid-operation rates, against the exposure-matched continued-SFT control and the two references.

| Arm                       | Dev success /9 | Unseen success /72 | Unseen invalid rate |
|---------------------------|----------------|--------------------|---------------------|
| DAgger iteration 2        | 3/9            | 1/72               | 0.316               |
| Continued SFT iteration 2 | 2/9            | 1/72               | 0.175               |
| Original process SFT      | 1/9            | 0/72               | 0.364               |
| Random-valid              | 6/9            | 45/72              | 0.000               |
| Exact reference           | 9/9            | 72/72              | 0.000               |

**Generalization and robustness v2.** The reduced v2 protocol admitted a uniform cheapest-prefix sample of 5 variants per family over 5 perturbation families (25 of 93 eligible semantics-preserving variants, membership frozen before execution) and evaluated the learned additive-best-first adapters. Learned adapters succeeded on 128/150 episodes versus 0/150 for the pretrained base, where every base episode is a one-call invalid termination (150 decisions total). By family, learned success was 29/30 on object-renaming, 27/30 on name-compression, 27/30 on render-restyle, 26/30 on scale-up, and 19/30 on shifted-init. By modality the split was text 44/50, multimodal 43/50, visual 41/50. By algorithm it was greedy 67/75 and weighted A\* 61/75. All 22 learned failures are single-invalid-operation terminations, 11 of them in shifted-init. Controls saturate (random-valid 750/750, exact 150/150) and remain bounds, not learned ability. The lossy name-compression stratum is reported separately and never pooled. All 5 admitted variants are text-lossy, 3 multimodal-lossy, and 0 visual-lossy, with learned success on that stratum of text 9/10, visual 8/10, multimodal 10/10. Family strata hold 5 problems each, so family-level rates are descriptive only. The full-scope v1 generalization branch was withdrawn after its terminal VALID.STOP receipt (closed 2026-09-21) and will not be executed.

**DAgger.** Two DAgger iterations (Ross et al., 2011) on BFS were compared against an exposure-matched continued-SFT control on 3 development and 24 unseen modality-task pairs (Table 8). On the unseen panel, DAgger and continued SFT each achieve 1/72 invariant-valid successes (paired over whole problems, one DAgger win, one loss, 70 ties), both far below random-valid at 45/72 and exact execution at 72/72, while the original process-SFT checkpoint achieves 0/72. The result is null. DAgger also carries the higher unseen invalid-operation rate of the two iterated arms (0.316 versus 0.175). Final cumulative unique training corrections were text 201, visual 203, multimodal 181, accumulated over 12 training cells of 512 records and 16 optimizer updates per cell-iteration.

**Model-generated successors.** Replacing trusted successors with model-generated dynamic-state successors collapses goal success from 45/45 to 1/45 on the evaluated panel, with zero trusted-state substitutions in the trusted arm. The single success is the one multimodal unseen cell. The trained successor model is exactly correct on 65/109 retained raw predictions, with 93 schema-valid and 66 effect-valid, and downstream failure kinds are dominated by exact state-identity and effect failures (27 effect, 16 schema, 1 static-context) rather than schema failures. Collection-time verification of the untrained exact-successor accepted 119/1,536 predictions (text 43, visual 52, multimodal 24). Coverage is partial by design. The frozen outcome-blind admission accepted a 12-task fallback panel after the full 24-task unseen panel was priced at 147.66 GPU-hours against the 64 GPU-hour branch cap, so full unseen coverage remains missing.

**Curriculum by modality.** Nine training cells (three modalities under staged, shuffled, and mixed ordering, all under greedy additive best-first and trained fresh from the same base on identical record sets) were evaluated on 27 unseen whole problems. Successes were 24/24/25 (staged/shuffled/mixed) in text-state, 24/27/26 in visual-state, and 25/27/27 in multimodal-state. A

Table 9: Zero-shot transfer accuracy on frozen benchmark subsets (FOLIO 200, GSM8K 200, HumanEval 164). Adapter rows follow the transfer ledger names: `bfs`, `iw` (width-based best-first, BFWS), `astar_w3`, and `astar_greedy`, each in text (t), visual (v), and multimodal (m) training cells. No comparison against the base survives Holm correction (0/36, minimum adjusted  $p$  0.8156).

| Adapter                             | FOLIO                 | GSM8K                 | HumanEval             |
|-------------------------------------|-----------------------|-----------------------|-----------------------|
| Base                                | 0.600                 | 0.955                 | 0.780                 |
| <code>bfs t / v / m</code>          | 0.600 / 0.605 / 0.605 | 0.950 / 0.945 / 0.965 | 0.793 / 0.799 / 0.780 |
| <code>iw t / v / m</code>           | 0.660 / 0.645 / 0.645 | 0.960 / 0.965 / 0.960 | 0.768 / 0.762 / 0.756 |
| <code>astar_w3 t / v / m</code>     | 0.590 / 0.575 / 0.590 | 0.950 / 0.965 / 0.945 | 0.817 / 0.799 / 0.774 |
| <code>astar_greedy t / v / m</code> | 0.590 / 0.590 / 0.590 | 0.950 / 0.955 / 0.950 | 0.799 / 0.811 / 0.793 |

sequential-SFT comparator reached 24/27, 25/27, and 24/27 in text, visual, and multimodal cells respectively, the base stayed at 0/27, and random-valid and exact execution saturated at 27/27 in every modality. All four paired interaction intervals include zero:  $-0.111 [-0.259, +0.037]$  and  $-0.074 [-0.259, +0.074]$  for the visual and multimodal staged-minus-shuffled contrasts against text, and  $-0.074 [-0.259, +0.111]$  and  $-0.037 [-0.185, +0.111]$  for the corresponding mixed-minus-shuffled contrasts. No material ordering-by-modality interaction emerged. Because both references saturate and the base sits at floor, ceiling and floor effects bound the observable differences.

**Transfer.** All 12 verified planning adapters plus the base were evaluated zero-shot on frozen subsets of FOLIO (Han et al., 2022) (200 items), GSM8K (Cobbe et al., 2021) (200), and HumanEval (Chen et al., 2021) (164). Table 9 gives the per-adapter accuracies. The paired verdict uses McNemar exact comparisons (McNemar, 1947) of each adapter against the per-example base predictions, Holm-corrected across all 36 comparisons (Holm, 1979). None survives correction. The minimum adjusted  $p$  is 0.8156, and the smallest raw  $p$  is 0.0227 (FOLIO with the text width-based adapter, at +12 examples). Leakage screening over the frozen 8-gram window finds 7/564 benchmark items sharing only trivial numeric 8-grams with the planning corpus and zero content overlap. The transfer result is a null.

**Second backbone.** Two reduced-scope cells trained and evaluated InternVL3.5-8B (Wang et al., 2025). Each used a frozen 12-task key-cell panel selected by lowest reference decision count before any model outcome, with one seed-17 training run per modality cell. The earlier cell (protocol v2) ran BFS, and neither arm reached a goal. The pretrained base scored 0/36 at 1 decision per episode (36 total), process SFT scored 0/36 at 1 to 4 decisions (59 total), and the reused BFS controls scored random-valid 54/72 (18/24 per modality, oracle-assisted) and exact 72/72. The replication cell (protocol v3, stress test R3) ran greedy additive best-first with the same backbone, recipe, and panel shape. Process SFT succeeds on 11/12 text-state, 10/12 visual-state, and 10/12 multimodal-state problems (31/36), while the base scores 0/36 with one invalid call per episode. Process-SFT invalid-operation rates are 0.00546, 0.01220, and 0.01258, and decision usage is 183, 164, and 159 of a 390-call allowance (text, visual, multimodal). Random-valid and exact reference both score 24/24 per modality. These control counts hold two entries per panel task, and both entries of a task come from its one reused, hash-pinned greedy baseline control episode per condition and modality. Paired whole-problem contrasts (seed 1729, 10,000 resamples) give SFT minus base  $+0.917 [+0.750, +1.000]$  in text-state and  $+0.833 [+0.583, +1.000]$  in both visual-state and multimodal-state. SFT minus random-valid is  $-0.083 [-0.250, +0.000]$ ,  $-0.167 [-0.417, +0.000]$ , and  $-0.167 [-0.417, +0.000]$  in the same order. Coverage is complete, with 72/72 model episodes and 72/72 comparators replayed. Learned success under the enumeration contract appears only in the additive cells, where the identity audit shows no decision to make (Section 5.2), and the InternVL v3 cell replicates that contract fluency (31/36 versus 0/36 base, below random-valid 24/24 per modality). The earlier total null on this backbone is consistent with an algorithm (choice-requirement) difference rather than a backbone null. The cross-backbone contrast against the pinned Qwen3-VL process-SFT rate is descriptive rather than paired, because the InternVL panel is the key-cell subset while the Qwen rate is the pinned full-panel aggregate. It is  $+0.000 [-0.167, +0.083]$  in text-state,  $-0.083 [-0.333, +0.083]$  in visual-state, and  $-0.042 [-0.292, +0.125]$  in multimodal-state. InternVL3.5-8B shares the Qwen3-8B LLM family (Yang et al., 2025), so the comparison differs in the vision tower, connector, image tokenization, and multimodal recipe, and it is reported as an architecture-difference limitation. The full-scope v1 second-backbone branch was withdrawn

after its terminal `VALID_STOP` receipt (closed 2026-09-21) and will not be executed, and its gate is retained as history.

## N RESULTS DETAIL: FAILURE-MECHANISM CALIBRATION

This appendix localizes where each arm class fails, for readers asking why learned success is absent under BFS and BFWs.

A frozen CPU-only calibration mined 2,910 unique replay-verified episodes across the five executed model branches and classified all 905 failures with zero unclassified. The protocol was frozen before any outcomes were computed, and the analysis is descriptive. Three arm-level profiles emerge. The base policy fails in 447 of 447 episodes, always on the first decision. Of these, 379 emit a malformed output that never reaches the operation checker, and 67 issue rejected BFS frontier operations. Learned arms fail in 407 of 906 episodes, and the dominant learned failure kind is `other_invariant_violation` with 229 occurrences (56.3% of learned failures), namely parsed, source-correct, applicable operations that the runtime rejects because they violate the declared search invariant. The remaining learned failures are 72 malformed outputs, 56 inapplicable grounded actions, 27 effect failures, 16 schema failures, and smaller kinds. First failures cluster early, at decision indices 0, 1, and 2 with counts 137, 88, and 46. The oracle control fails 51 times, all late and all by expansion-budget exhaustion (buckets 5–8, 9–16, 17–32, and 33+ hold 9, 9, 3, and 30). Table 1 gives the rejection taxonomy.

Every baseline process-SFT episode under BFS or BFWs fails (144/144). The DAGger-iteration BFS arms fail 155/162, while the learned additive best-first families fail 56 of 546 episodes (10.3%). This pattern matches the primary matrix (Section 5.6). The calibration is descriptive. After process supervision, most learned failures (229/407) are parsed, source-correct operations rejected by the search invariant, and 72/407 remain malformed outputs. Read with the horizon profile, the learned policy fails early on invariant grounds, whereas the oracle control, which always emits valid operations, fails only when its expansion budget runs out late in the episode.

## O RESULTS DETAIL: OBSERVATION-CORRUPTION CHANNEL TESTS

This appendix reports the observation-corruption tests, which show which observation channel the trained enumeration-contract controller relies on.

The observation-corruption suite evaluated frozen checkpoints, with no retraining, on a prospectively frozen set of  $k=9$  admitted tasks under four deterministic corruption families applied at observation time. Clean learned success on these tasks is 0.917 (text-state 0.944, visual-state 0.889, multimodal-state 0.889, with 9 tasks  $\times$  2 algorithms per cell). Table 10 gives the paired whole-problem deltas. Copyability-preserving text shuffling lowers success by up to  $-0.944$   $[-1.000, -0.833]$ , and the multimodal arm retains at most 0.167 success under shuffled text. Masking instead destroys the output contract. Visual-channel corruption is not material. Blanking or degrading the rendered state costs at most  $-0.111$   $[-0.278, +0.000]$ , and every visual interval includes zero. The multimodal channel-isolation contrast, paired by task and algorithm ( $n=18$ ), is  $-0.806$  for text-channel corruption versus  $-0.111$  for visual-channel corruption, giving text minus visual  $-0.694$  with 95% interval  $[-0.889, -0.472]$ , which is material under the frozen rule. This contrast pools the output-contract-destruction (masked) regime. Controls behave as the contract predicts. The corrupted base stays at floor (0/144, all first-decision invalid), random-valid succeeds 72/72 on the new frozen seeds, so per the saturation rule no structural-advantage claim is made, and exact execution remains at 18/18.

A decomposition of the 72 published text-corruption episodes (stress test R2) separates two regimes that the success deltas pool together. Under text masking, every episode of the text-state and multimodal-state cells is all-invalid (18/18 in each, 18 invalid operations over 18 decisions). Each emitted operation fails the schema, so the policy never enters the search. This is output-contract destruction, not information loss. Copyability-preserving text shuffling behaves differently, because valid operations keep flowing while the search fails. In the text-state cell 16/18 episodes are partial-valid, 2/18 are all-invalid, and none reaches the goal (18 invalid operations over 49 decisions). In the multimodal-state cell 15/18 are partial-valid, none is all-invalid, and 3/18 reach the goal (15 in-

Table 10: Observation corruption on frozen checkpoints ( $k=9$  admitted tasks, paired whole-problem deltas against clean runs, strata below 8 descriptive-only under the frozen rule). “Material” is the frozen protocol’s interval-based verdict.

| Corruption      | Arm        | Corrupted | Clean | $\Delta$ | 95% interval       | Material |
|-----------------|------------|-----------|-------|----------|--------------------|----------|
| Text-masked     | text       | 0.000     | 0.944 | -0.944   | $[-1.000, -0.833]$ | yes      |
| Text-masked     | multimodal | 0.000     | 0.889 | -0.889   | $[-1.000, -0.722]$ | yes      |
| Text-shuffled   | text       | 0.000     | 0.944 | -0.944   | $[-1.000, -0.833]$ | yes      |
| Text-shuffled   | multimodal | 0.167     | 0.889 | -0.722   | $[-0.889, -0.500]$ | yes      |
| Visual-blank    | visual     | 0.833     | 0.889 | -0.056   | $[-0.167, +0.000]$ | no       |
| Visual-blank    | multimodal | 0.778     | 0.889 | -0.111   | $[-0.278, +0.000]$ | no       |
| Visual-degraded | visual     | 0.833     | 0.889 | -0.056   | $[-0.167, +0.000]$ | no       |
| Visual-degraded | multimodal | 0.778     | 0.889 | -0.111   | $[-0.278, +0.000]$ | no       |

Table 11: Menu manipulation on the frozen native adapters (both arms, two additive algorithms). Invalid operations are counted over emitted operations. The distractor pick-rate is counted over decisions with injected distractors.

| Family               | Success | Invalid ops | Distractor picks |
|----------------------|---------|-------------|------------------|
| Menu permutation     | 36/36   | 0/512       | n/a              |
| Distractor injection | 0/36    | 36/60       | 36/60 (0.6)      |

valid operations over 105 decisions). This partial-valid pattern is the information-loss signature. A new cross cell applies text masking to the frozen visual-state arm, whose nominal state channel is the image, and it scores 0/18 with every episode all-invalid. Masking the text scaffold destroys the output contract even when images are the nominal state channel.

The language-channel claim survives only in its shuffled form. The masked catastrophe in Table 10 is mostly a loss of the copyable output format rather than a loss of state information. Shuffling the text leaves operations copyable and still stops the search, so under the frozen observation contract the textual serialization carries decision-relevant information for these adapters. Removing or degrading the visual channel barely moves success, including on the visual-state arm whose only rendered-semantics channel was corrupted. Sections P and Q probe the grounded candidate menu directly. This is a channel finding about the trained controller under the declared adapters, not evidence about intrinsic modality value.

## P RESULTS DETAIL: MENU-MANIPULATION RECEIPTS

Menu manipulation (stress test R1) probes how the trained policy responds to changes in the grounded menu. It runs on the frozen image-only adapters of both native arms (the Image-Only Native Arms appendix, Appendix Q), with no retraining, under two families. Menu permutation reorders the candidate menu. Distractor injection adds  $k=3$  schema-valid but inapplicable actions to it. Table 11 reports the 72 episodes, all independently replayed within the 170/170 stress-evaluation episodes.

On the frozen native adapters (stress test R1, 72 replayed episodes), menu permutation leaves 36/36 successes and 0/512 invalid operations, which is uninformative about order under the submission-order invariance of Section 5.2. Injecting  $k=3$  schema-valid but inapplicable actions collapses success to 0/36, with 36/60 invalid operations and a distractor pick-rate of 0.6 against a uniform-picking expectation of 0.592 to 0.625, and every episode ends on a distractor pick, 20/36 on the first decision. The receipt therefore shows that the policy does not reject schema-valid inapplicable menu entries (pick-rate at chance), and these out-of-distribution distractors never appeared in training menus. Content-reading and position habits both predict a pick-rate near 0.6, so this receipt does not separate them, and the discriminating fixed-versus-random-position test remains unrun.

## Q RESULTS DETAIL: IMAGE-ONLY NATIVE ARMS

This appendix reports the image-only native arms, which test whether removing the textual scaffold changes the enumeration-contract results.

The native-arms follow-up (contract v2) trained two image-only observation arms. In `visual-nomem-state` (nomem) and `visual-seq-state` (seq), state facts arrive only as rendered images, and seq adds a window of  $K=8$  history frames. Neither arm receives textual search memory,  $g/h$ /priority scalars, candidate scores, or membership flags. The grounded, unscored candidate menu is still presented. The v1 admission gate measured pretrained-base format compliance and is retained as history. The v2 amendment, frozen before any trained-arm outcome, replaced it with a pre-registered post-training smoke gate, which both arms passed with 62/62 schema-valid grounded calls against a threshold of 0.5. Four adapters (two arms by two additive best-first algorithms) used the frozen recipe with 512 records, 16 updates, and seed 17. All 216/216 new bindings were recorded and independently replayed, together with 72/72 reused replay-verified comparators. The window spent 3.2086 of 12 GPU-hours.

On the clean panel, both arms nominally succeed on 18/18 with zero invalid operations, against 16/18 for the published frozen `visual-state` arm. The pretrained base scores 0/36 under both new contracts and exact reference scores 18/18. Random-valid is at ceiling on its five frozen seeds, so under the saturation rule no structural-advantage claim is made. Paired whole-problem contrasts ( $n=18$ , bootstrap seed 61813, 10,000 resamples, 95%) give nomem minus `visual-state` +0.111 [0.000, +0.278] and seq minus `visual-state` +0.111 [0.000, +0.278], neither material. The history window changes nothing on these tasks (seq minus nomem 0.000 [0.000, 0.000]). Learned minus pretrained base is +1.000 [+1.000, +1.000] for each arm, which is material.

The pre-registered corruption payoff is a null. Under seq the images are the only state and history channel, yet blanking every attached image gives a seq corrupted-minus-clean contrast of 0.000 [0.000, 0.000], and degrading every image gives 0.000 [0.000, 0.000]. The images-only hypothesis is therefore answered in the negative on this panel. As a post-hoc interpretation, the grounded unscored candidate menu is the remaining channel that could carry the signal, since neither the rendered scenes nor the removed scalar scaffold is necessary once the menu is present. On these additive cells the identity audit (Section 5.2) makes any listed candidate a correct emission, so the null is equally consistent with a policy that uses no state information.

The degraded-strain test (stress test R4) evaluated the same frozen native adapters on the two generalization families where the full-scaffold adapters had degraded. The learned adapters succeed on 10/10 shifted-init and 10/10 scale-up episodes for both arms, and the pretrained base scores 0/20 on each strain (0/40 in total). The published full-scaffold rates on the same variants are 19/30 for shifted-init and 26/30 for scale-up (Secondary Panels appendix, Appendix M). The two rates come from different episode sets and are not paired. The reduced contract sits at ceiling where the full scaffold degraded. A post-hoc, unpaired reading, with 10 visual-only episodes per arm against 30 full-scaffold episodes pooled over three modalities, is consistent with the reduced contract removing scalar operands whose emission the full-scaffold adapters failed, and it does not measure whether that text carried signal, since any listed candidate succeeds on these cells.

## R RESULTS DETAIL: CHOICE-FRONTIER AND COMPARATOR-ZOO RECEIPTS

This appendix holds the receipts of the first choice-frontier measurement and the comparator zoo on the reused panel, for readers checking the contract’s first identity-audit pass.

The training corpus was derived on CPU by replaying the frozen exact traces through the new contract and verified expansion-for-expansion against them (58 episodes bit-exact). It covers 29 training tasks and holds 1,046 records (512 training plus 11 diagnostic per algorithm) with 3,544 scenes. Every corpus episode carries at least one decision whose menu offers two or more states (up to 315 such decisions in one episode, menus of up to 222 states). Two adapters, one per additive algorithm, were trained with the frozen recipe at seed 17 (16 updates, 512 records each), and the smoke gate accepted 40/40 model calls. All 144/144 panel episodes were independently replayed with complete coverage. The window spent 1.069 of 12 GPU-hours (training 0.81, smoke 0.052, model evaluation 0.208), and corpus and controls ran on CPU.

Paired bootstrap contrasts (seed 61813, 10,000 resamples) give random-valid minus exact reference  $-0.644$   $[-0.856, -0.422]$ , material, so the controls separate under this contract. The learned adapter minus pretrained base is +0.389 [+0.167, +0.611], material, with zero invalid operations, a contract-fluency gain. The learned adapter minus random-valid is +0.033  $[-0.033, +0.133]$ , not

Table 12: First measurement under the choice-frontier contract on the frozen 9-task panel, with 18 cells (9 tasks  $\times$  2 additive algorithms as repeated measures) and 144 episodes independently replayed. Exact reference solves by construction of the budget and is the imitation target. Not comparable with enumeration-contract rows.

| Condition                     | Goal rate | Failures                 | Invalid ops |
|-------------------------------|-----------|--------------------------|-------------|
| Exact reference               | 18/18     | none                     | 0           |
| Learned adapter               | 7/18      | 11 budget exhausted      | 0           |
| Pretrained base               | 0/18      | 18 invalid at first call | 18          |
| Random-valid (5 frozen seeds) | 32/90     | 58 budget exhausted      | 0           |

Table 13: Exploratory pre-registered metric set at the frozen shared  $2\times$  cap (9 panel tasks  $\times$  2 additive settings). M1 is the task-clustered solve-versus-budget AUC over  $m \in \{1, 1.25, 1.5, 1.75, 2\}$  (9-task bootstrap, seed 133, 10,000 draws). M2 is the solved-only task-weighted geometric expansion overhead  $\rho$  and M3 the solved-only task-weighted cost ratio  $\kappa$ , both selection-conditioned. The  $3\times$  and  $4\times$  columns are additional CPU controls, not M1. Selectors read privileged information that the model does not see.

| Arm                     | M1 AUC [95% CI]      | $2\times$ solved | $3\times$ | $4\times$ | M2 $\rho$ | M3 $\kappa$ |
|-------------------------|----------------------|------------------|-----------|-----------|-----------|-------------|
| Learned adapter         | .1042 [.0278, .2014] | 7/18             | n/a       | n/a       | 1.432     | 1.025       |
| Pretrained base         | 0 [0, 0]             | 0/18             | n/a       | n/a       | undefined | undefined   |
| Random-valid            | .1389 [.0250, .2722] | 32/90            | 42/90     | 50/90     | 1.407     | 1.000       |
| Exact reference         | .8472 [.7917, .8750] | 18/18            | n/a       | n/a       | 1.000     | 1.000       |
| bfs-order               | .0694 [0, .1528]     | 6/18             | 8/18      | 10/18     | 1.613     | 1.000       |
| Novelty-first selection | .1250 [0, .2778]     | 6/18             | 8/18      | 12/18     | 1.452     | 1.000       |
| worst-first             | .0556 [0, .1389]     | 4/18             | 6/18      | 8/18      | 1.549     | 1.050       |

material. The 18 cells are 9 tasks with two algorithms as repeated measures, and that last contrast rests entirely on one task (storage), the only one where random-valid’s seed frequency lies strictly between 0 and 1. That falls below the frozen rule that strata under eight cells are descriptive only, so the contrast is descriptive. The evidence supports no learning-to-plan claim for the adapter, and no equivalence claim with random-valid, since no equivalence margin was pre-registered.

The control expansion sequences coincide across the two algorithms on every panel task. Post-hoc task-clustered intervals with the same seed and resample count are  $[0.000, +0.100]$  for learned minus random-valid and  $[-0.889, -0.333]$  for random-valid minus exact reference.

**Comparator zoo.** The CPU-only comparator zoo evaluates this measurement against three deterministic selectors at the same  $2\times$  cap (bfs-order, novelty-first selection, and negative control worst-first). Its metric set was pre-registered before comparator episodes, but stored choice-frontier episodes enter post-hoc. The selectors act on the additive frontier, not native BFS or BFWS, and read privileged scalars or PDDL atoms while the model sees only scenes, so no information-matched learned-versus-rule claim follows.

On M1 the learned adapter falls between bfs-order and random-valid and below novelty-first selection (Table 13, Figure 3). No comparator between random-valid and exact exists on this panel, so the zoo bounds the adapter only against weaker-than-random rules and cannot calibrate choice quality. Post-hoc M4 finds 17.4% heap-head agreement against 26.9% chance and 54.5% last-menu-label emissions, and the task-clustered interval of the agreement gap includes zero (Counterfactual appendix, Appendix S), with no last-label comparator and no equivalence claim. Across the fifteen-test Holm family no comparison supports positive learned Pareto dominance, with three co-solved tests undefined. Non-significance is neither equivalence nor a generalized negative capability claim, and the one-task M5 subset is descriptive only.

**Zoo receipts.** Random-valid is the zoo’s uniform-choice reference and exact reference the imitation target. Random-valid and the three selectors also ran at  $3\times$  and  $4\times$  caps, outside the primary metric. Because the CPU selectors operate on the additive frontier, novelty-first selection does not reproduce BFWS pruning. Selector episodes replay the frozen model input, randomized menu, actions, admission, and budget, but do not generate or replay image view bindings. All 432/432 zoo bindings were observed and independently replayed, with 342 new CPU episodes, 90 original  $2\times$  random-valid controls reused, and none missing. The nine unit-cost optimal costs  $C^*$  come from one offline planner computation per task with replay-verified plans: storage 3; elevators, ferry, and 15puzzle-compact 4; depot, towers-of-hanoi, and visitall 5; blocksworld and 15puzzle-expanded 6.

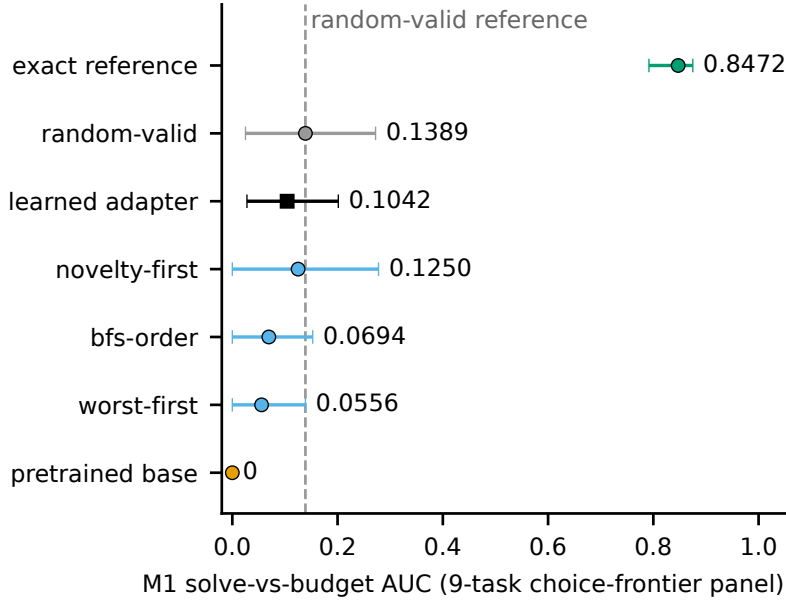


Figure 3: Solve-versus-budget AUC (M1) on the frozen 9-task choice-frontier panel at multipliers 1 to 2. Exact reference .8472, random-valid uniform-choice reference .1389, learned adapter .1042 [.0278, .2014], novelty-first .1250, bfs-order .0694, worst-first .0556, pretrained base 0. Intervals of all non-exact arms overlap and the Holm family licenses no ordering. Under the enumeration contract, the BFS/BFWS random-valid rows (15/24 and 17/24 versus exact 24/24) already registered choice.

The selectors are `bfs-order` (minimum frontier  $g$ , then earliest serial), novelty-first selection (lowest width-1 then width-2 novelty over positive PDDL atoms), and `worst-first` as a negative control.

Exploratory post-hoc M4 on the stored episodes shows the learned adapter selecting the teacher’s heap-head state on 23/132 multi-option decisions (17.4%) against an on-policy chance rate of 26.9%, and emitting the last menu label on 72/132 (54.5%). Decisions cluster within 9 tasks, and the task-clustered intervals in the Counterfactual appendix (Appendix S) show that the agreement gap does not survive that clustering while the last-label excess does. The post-hoc choice-frontier diagnostic records a first-label rate of 2.3% and a teacher-corpus last-label rate of 12.0% (chance 10.4%, 978 records). Menus are permuted per decision with a seed, and no last-label comparator was run, so these post-hoc rates do not establish a position habit. M5 restricts to tasks where random-valid solves at a  $2\times$  rate strictly between 0 and 1. That subset holds one task (storage-compact-919000, random-valid rate .2), so M5 is descriptive only. On it the learned adapter’s M1 AUC is .3125, against .075 for random-valid and .625 for exact reference.

The choice-frontier contrasts use percentile 95% intervals with the frozen bootstrap resampling cells. Random-valid’s seed frequency on storage is 0.2. The final audit returns `CHOICE_SENSITIVE` and passes over 144 replayed episodes.

## S RESULTS DETAIL: SUBMISSION-ORDER COUNTERFACTUAL AND M4 INTERVALS

This appendix gives the submission-order counterfactual behind the enumeration-contract identity audit and the task-clustered intervals for the first adapter’s post-hoc diagnostics.

A CPU-only follow-up with a protocol committed before its outputs re-ran the identity audit on the same 48 additive pairs (24 tasks  $\times$  2 algorithms, seed 17). The executed rule assigns heap serials at expansion start from the sorted candidate order, and the counterfactual assigns them at application time in submission order. The audit first re-validated 288 stored episode counts and 96 raw-output sequences with no mismatch.

Table 14: Random-valid against exact reference on the 48 additive pairs under the two heap-serial rules.

| Heap-serial rule                  | Identical | Divergent |
|-----------------------------------|-----------|-----------|
| Sorted candidate order (executed) | 48        | 0         |
| Submission order (counterfactual) | 29        | 19        |

Table 15: Post-hoc M4 diagnostics for the first adapter on the earlier 9-task panel.

| Quantity                         | Pooled | Task-clustered [95% CI] |
|----------------------------------|--------|-------------------------|
| Heap-head agreement minus chance | -0.095 | -0.069 [-0.161, +0.032] |
| Last-label rate minus chance     | +0.276 | +0.277 [+0.126, +0.402] |

The stored verdict reads “PREDICTION\_FAILED: see preregistered\_prediction”. The executed rule stays identical as predicted, but 19/48 divergent pairs fall short of the pre-registered threshold of more than 24/48. First divergence falls between decisions 2 and 50 (median 6), and both algorithms on one ferry task flip the goal outcome. The 48/48 identity is therefore a property of the executed serial rule, not of the enumeration contract alone.

A post-hoc analysis recomputed the first adapter’s M4 diagnostics over its 18 stored episodes on the earlier panel (9 task clusters, 10,000 resamples, seed 133). Pooled over 132 multi-option decisions, heap-head agreement is 0.174 against a chance rate of 0.269, and the last-label rate is 0.545. The agreement gap’s interval includes zero while the last-label excess excludes it (Table 15), and M4 keeps its diagnostic, non-confirmatory status.

## T RESULTS DETAIL: CHOICE-FRONTIER VALIDATION PANEL AND LADDER

This appendix holds the validation panel’s construction, every arm’s receipts, and all adjacent ladder gaps behind the ladder test in Section 5.3.

The ladder protocol was committed before task generation and the panel membership before evaluation. Of 240 generated candidates (12 snapshots, seeds 935000 to 935019), 127 met the cost filter ( $8 \leq C^* \leq 20$ ) and 34 passed controls-only screening, which admits tasks on which random-valid is discriminative. The freeze rule sorts admitted tasks by domain and seed, keeps at most two per domain, and takes the first 12 (Table 16).

All 696/696 CPU episodes and the 48 first-adapter model episodes were independently replayed. PASS requires all four adjacent paired task-cluster lower bounds to be positive, with arms averaged over seeds within each task and algorithm, then over algorithms. The gaps are +0.082 [+0.051, +0.111] (exact reference minus 0.25), +0.190 [+0.126, +0.252] (0.25 minus 0.50), +0.256 [+0.184, +0.339] (0.50 minus 0.75), and +0.326 [+0.233, +0.422] (0.75 minus random-valid).

Three deviations are documented. Generation reuses the frozen candidate generator, which reproduces all 12 snapshots byte-for-byte although domain generators take the seed differently. A first concurrent generation run corrupted the blocksworld outputs through a shared temporary file, so it was discarded, all candidates were regenerated serially and reproduced, and the deviation was disclosed before evaluation. Only the two additive native references seed the pre-rendered views, rather than four algorithms, because native BFS exceeds the 128-expansion reference ceiling on these tasks, and unseen states are rendered live.

## U RESULTS DETAIL: SCALED ADAPTER AND SEED REPLICATION

This appendix gives the adapter’s protocol, corpus, recipe, and per-seed results behind Section 5.4, with the held-out panels, the DAgger ablation, and the zero-shot policy as subsections.

The protocol was frozen before task generation, the corpus before training, and the schedule before launch.  $D$  is the mean over training seeds of learned M1 minus random-valid M1, with a task-cluster interval (10,000 draws, seed 133). POSITIVE requires a positive lower bound,  $D \geq 0.05$ , and a positive estimate at every seed, and the equivalence margin is  $\pm 0.05$ . Controls are reused from the ladder test.

Table 16: The frozen validation panel (12 tasks, 7 domains).  $R$  is the uncapped exact-reference expansion count (greedy / w3), and Screen counts random-valid goals over 10 screening episodes.

| Task                   | $C^*$ | $R$     | Screen |
|------------------------|-------|---------|--------|
| 15puzzle-935006        | 8     | 46 / 12 | 1      |
| blocksworld-935001     | 10    | 26 / 26 | 4      |
| blocksworld-935004     | 10    | 13 / 13 | 2      |
| depot-935005           | 10    | 14 / 15 | 1      |
| depot-935011           | 9     | 13 / 13 | 1      |
| elevators-935000       | 8     | 11 / 11 | 2      |
| elevators-935013       | 8     | 11 / 11 | 2      |
| grid-935016            | 8     | 9 / 9   | 4      |
| storage-935000         | 10    | 20 / 17 | 1      |
| storage-935001         | 8     | 18 / 15 | 2      |
| towers_of_hanoi-935000 | 13    | 24 / 25 | 1      |
| towers_of_hanoi-935002 | 15    | 24 / 26 | 2      |

Table 17: All arms on the validation panel. M1 carries a task-clustered 95% interval (12-task bootstrap, seed 133, 10,000 draws), and M2 and M3 are solved-only. Only the five ladder arms enter the pre-registered test. The descriptive selectors read privileged information, and the base emits no valid first choice under its one-call cap.

| Arm                      | M1 [95% CI]          | $2\times$ | M2 $\rho$ | M3 $\kappa$ |
|--------------------------|----------------------|-----------|-----------|-------------|
| Exact reference          | 0.875 [0.875, 0.875] | 1         | 1.0       | 1.021       |
| Exact- $\epsilon$ 0.25   | 0.793 [0.764, 0.824] | 0.992     | 1.089     | 1.024       |
| Exact- $\epsilon$ 0.50   | 0.603 [0.528, 0.670] | 0.95      | 1.246     | 1.032       |
| Exact- $\epsilon$ 0.75   | 0.347 [0.252, 0.445] | 0.717     | 1.374     | 1.023       |
| Random-valid             | 0.021 [0.007, 0.038] | 0.083     | 1.692     | 1.042       |
| hadd-greedy              | 0.882 [0.774, 0.955] | 0.958     | 0.962     | 1.083       |
| bfs-order                | 0.000 [0.000, 0.000] | 0         | undefined | undefined   |
| Novelty-first selection  | 0.021 [0.000, 0.052] | 0.167     | 1.909     | 1.000       |
| worst-first              | 0.000 [0.000, 0.000] | 0         | undefined | undefined   |
| First adapter            | 0.026 [0.010, 0.042] | 0.208     | 1.810     | 1.050       |
| Scaled adapter (seed 17) | 0.349 [0.172, 0.531] | 0.583     | 1.326     | 1.140       |
| Pretrained base          | 0.000 [0.000, 0.000] | 0         | undefined | undefined   |

Of 2,400 serially generated training tasks, 1,182 remained eligible after excluding overlap with every evaluation panel and deduplicating, and the seed-major walk reached 2048 records per algorithm from 236 tasks in 12 domains. Two reseeded menu orders per record give 8,192 samples, with the target last in 17.58% of samples against a chance rate of 17.58%.

The recipe matches the first adapter’s except for corpus, augmentation, sample order, and seed (LoRA rank 64, alpha 128, 128 steps, seed 17). The smoke gate accepted 34/34 calls, and all 24/24 learned episodes, capped at  $2R$ , replayed with no mismatch and no invalid operation. For seed 17, the  $2\times$  solve fraction is 0.583, M2  $\rho$  1.326, and M3  $\kappa$  1.140. Over 691 multi-option decisions heap-head agreement is 0.210 against chance 0.125, the last-label rate is 0.205, and the first-label rate is 0.090.

Four choices are documented. The scope edit to one training seed left seed variance unmeasured in that run. A later pre-registered replication trained seeds 29 and 71 with the byte-identical recipe and the sample order pinned to the seed-17 order, so seeds change only the LoRA initialisation and dropout, and all 48 of its new episodes were independently replayed with no mismatch. The step budget matches the first adapter’s recipe. The seed-major corpus walk resolves an ambiguity in the protocol text. Both evaluation hooks first failed because the scheduler passes only the terminal path, and after a fix deriving the attempt directory from it, the audit re-ran on the same attempts and passed without affecting GPU work or episodes.

On the validation panel,  $D$  is  $+0.328$  [ $+0.155, +0.508$ ] for seed 17,  $+0.229$  [ $+0.108, +0.354$ ] for seed 29, and  $+0.297$  [ $+0.134, +0.466$ ] for seed 71. Per-seed M1 is 0.349, 0.250, and 0.318, with a between-seed standard deviation of 0.051 (descriptive). The last-label rate per seed is 0.205, 0.192, and 0.171 (chance about 0.13, descriptive), against 0.300 (chance 0.144) for the first adapter on the same panel. The pre-registered separation from the exact- $\epsilon$  0.75 rung is  $S = +0.002, -0.097$ , and  $-0.029$  per seed. Against the 0.50 rung the three-seed  $S$  is  $-0.298$  [ $-0.468, -0.129$ ] (descriptive). Table 20 sets these values beside the held-out and unscreened panels.

Table 18: Per-task  $D$  (adapter M1 minus random-valid M1) for seed 17 on the validation panel. The pre-registered three-seed endpoint is  $D_3 = +0.285 [+0.140, +0.433]$  (Table 20).

| Task               | $D$     | Task                   | $D$    |
|--------------------|---------|------------------------|--------|
| 15puzzle-935006    | -0.0625 | elevators-935013       | 0.6875 |
| blocksworld-935001 | 0.0     | grid-935016            | 0.3625 |
| blocksworld-935004 | 0.0     | storage-935000         | 0.7375 |
| depot-935005       | 0.0625  | storage-935001         | 0.7    |
| depot-935011       | 0.7125  | towers_of_hanoi-935000 | 0.2375 |
| elevators-935000   | 0.5     | towers_of_hanoi-935002 | 0.0    |

Table 19: M1 with task-clustered 95% intervals on the validation panel (12 tasks), the held-out panel P2 (11 tasks), and the unscreened panel P2u (12 tasks). All entries are descriptive. Ladder rungs and hadd-greedy read privileged information and are not realisable policies.

| Arm                      | Validation (12)      | P2 (11)              | P2u (12)             |
|--------------------------|----------------------|----------------------|----------------------|
| Exact reference          | 0.875 [0.875, 0.875] | 0.875 [0.875, 0.875] | 0.875 [0.875, 0.875] |
| Exact- $\epsilon$ 0.25   | 0.793 [0.764, 0.824] | 0.745 [0.685, 0.805] | 0.678 [0.632, 0.721] |
| Exact- $\epsilon$ 0.50   | 0.603 [0.528, 0.670] | 0.507 [0.445, 0.566] | 0.397 [0.331, 0.478] |
| Exact- $\epsilon$ 0.75   | 0.347 [0.252, 0.445] | 0.205 [0.145, 0.269] | 0.149 [0.091, 0.227] |
| Random-valid             | 0.021 [0.007, 0.038] | 0.016 [0.000, 0.048] | 0.002 [0.000, 0.006] |
| Pretrained base          | 0.000 [0.000, 0.000] | 0.000 [0.000, 0.000] | 0.000 [0.000, 0.000] |
| Adapter, seed 17         | 0.349 [0.172, 0.531] | 0.506 [0.313, 0.699] | 0.198 [0.031, 0.396] |
| Adapter, seed 29         | 0.250 [0.125, 0.375] | 0.284 [0.119, 0.489] | 0.120 [0.026, 0.224] |
| Adapter, seed 71         | 0.318 [0.151, 0.490] | 0.506 [0.295, 0.705] | 0.135 [0.031, 0.250] |
| Adapter, three-seed mean | 0.306 [0.158, 0.455] | 0.432 [0.273, 0.602] | 0.151 [0.030, 0.288] |
| hadd-greedy (privileged) | 0.882 [0.774, 0.955] | 0.928 [0.883, 0.966] | 0.906 [0.883, 0.933] |

## U.1 HELD-OUT PANELS

Both fresh panels come from 2,400 candidates generated from seeds 955000–955199. Overlap with earlier panels removed 1,168 candidates, duplicates 205, goal at the initial state 31, and  $C^*$  out of range 536, which left 460. P2 applies the validation panel’s generation,  $C^*$ , and screening rules. A candidate is kept when  $8 \leq C^* \leq 20$  and random-valid reaches the goal in 1 to 9 of its 10 screening episodes, and the freeze walk, sorted by domain and seed, takes at most 2 tasks per domain up to 12 tasks, with a minimum of 8 tasks from 5 domains. P2 froze with 11 tasks from 8 domains because only 11 admitted tasks were available under the per-domain limit. It met the minimum, so every P2 test is confirmatory. P2u takes kept candidates outside P2 on which the exact reference reaches the goal under both algorithms, without consulting the random-valid screen, at most 2 per domain, giving 12 tasks from 7 domains. All 12 P2u tasks happen to have 0/10 random-valid goals in screening. All 638/638 P2 and 696/696 P2u zoo episodes were independently replayed, with none missing.

The first freeze, from the original 40-seed pool 955000–955039, admitted 6 tasks from 5 domains and took the protocol’s blocked exit. Only controls-only screening had run, and no evaluation episode existed. An amendment then extended the seeds to 955000–955199. It kept the 480 earlier candidates unchanged and generated the new seeds 955040–955199 serially after them. The generation mechanism, the exclusion sources, the duplicate check, the  $C^*$  rule,  $R_t$ , the screen, the freeze walk, and the P2u rule stayed unchanged, and both panels were frozen from the combined pool before any evaluation episode.

Table 19 gives each arm’s M1 on the three panels. The four pre-registered adjacent gaps on P2 are  $+0.130 [+0.070, +0.190]$ ,  $+0.239 [+0.183, +0.290]$ ,  $+0.302 [+0.259, +0.340]$ , and  $+0.189 [+0.136, +0.248]$ , so the ladder test passes. On P2u the descriptive gaps are  $+0.197 [+0.154, +0.243]$ ,  $+0.281 [+0.191, +0.355]$ ,  $+0.248 [+0.209, +0.290]$ , and  $+0.147 [+0.088, +0.226]$ , and every lower bound is positive.

Table 20 gives  $D$  per seed and  $D_3$  with intervals on each panel and pooled over 35 tasks, and the separation  $S$  from the exact- $\epsilon$  rungs. Seed 29 has the lowest per-seed  $D$  on every panel, and on P2 its separation point from the 0.75 rung is  $+0.080$  (descriptive). No equivalence margin was pre-registered for  $S$ , so a non-separated  $S$  is not equivalence to the rung. The P2 separation rests on 11 task clusters, with 8 positive and 3 negative per-task differences, and percentile intervals can under-cover at this size.

Table 21 counts tasks by the sign of their three-seed difference and leaves out one domain at a time over the validation and held-out panels. Over the 35-task union 22 tasks are positive, 12 are zero, and 1 is negative. On P2u the eight zero tasks are both 15-puzzle, both blocksworld, both depot,

Table 20: Per-seed  $D$  and three-seed  $D_3$  (adapter M1 minus random-valid M1), and three-seed separation  $S$  (adapter M1 minus exact- $\epsilon$  rung M1), with paired task-clustered 95% intervals (10,000 draws, seed 133). Pre-registered entries are marked pre-reg. and the rest are descriptive.

|               | Validation (12)               | P2 (11)                     | P2u (12)                      | Pooled (35)                 |
|---------------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|
| $D$ , seed 17 | +0.328 [+0.155, +0.508]       | +0.490 [+0.294, +0.688]     | +0.196 [+0.031, +0.396]       | +0.334 [+0.223, +0.450]     |
| $D$ , seed 29 | +0.229 [+0.108, +0.354]       | +0.268 [+0.093, +0.477]     | +0.118 [+0.026, +0.220]       | +0.203 [+0.124, +0.290]     |
| $D$ , seed 71 | +0.297 [+0.134, +0.466]       | +0.490 [+0.276, +0.694]     | +0.133 [+0.031, +0.248]       | +0.301 [+0.200, +0.411]     |
| $D_3$         | +0.285 [+0.140, +0.433]       | +0.416 [+0.250, +0.591]     | +0.149 [+0.030, +0.283]       | +0.279 [+0.191, +0.375]     |
| $D_3$ verdict | POSITIVE (pre-reg.)           | POSITIVE (pre-reg.)         | POSITIVE (pre-reg.)           | POSITIVE (descriptive)      |
| $S$ vs 0.75   | -0.041 [-0.187, +0.102]       | +0.227 [+0.053, +0.403]     | +0.002 [-0.145, +0.157]       | +0.058 [-0.040, +0.156]     |
| $S$ verdict   | NOT_SEPARATED (pre-reg.)      | SEPARATED_ABOVE (pre-reg.)  | NOT_SEPARATED (descriptive)   | NOT_SEPARATED (descriptive) |
| $S$ vs 0.50   | -0.298 [-0.468, -0.129]       | -0.075 [-0.247, +0.089]     | -0.246 [-0.400, -0.082]       | -                           |
| $S$ verdict   | SEPARATED_BELOW (descriptive) | NOT_SEPARATED (descriptive) | SEPARATED_BELOW (descriptive) | -                           |

Table 21: Concentration of the three-seed difference (descriptive). Top, tasks by the sign of the per-task difference. Bottom,  $D_3$  over the validation panel and P2 with one domain left out, and the number of tasks left.

| Panel           | Positive | Zero             | Negative   |
|-----------------|----------|------------------|------------|
| Validation (12) | 9        | 2                | 1          |
| P2 (11)         | 9        | 2                | 0          |
| P2u (12)        | 4        | 8                | 0          |
| Union (35)      | 22       | 12               | 1          |
| Domain left out | $D_3$    | 95% interval     | Tasks left |
| 15-puzzle       | +0.384   | [+0.269, +0.501] | 21         |
| blocksworld     | +0.395   | [+0.269, +0.520] | 19         |
| depot           | +0.353   | [+0.224, +0.484] | 20         |
| elevators       | +0.305   | [+0.185, +0.430] | 20         |
| ferry           | +0.331   | [+0.212, +0.455] | 21         |
| grid            | +0.331   | [+0.204, +0.464] | 20         |
| storage         | +0.332   | [+0.209, +0.459] | 21         |
| Tower of Hanoi  | +0.384   | [+0.262, +0.507] | 20         |
| visitall        | +0.318   | [+0.211, +0.422] | 22         |

and both driverlog tasks, on which neither the adapter nor random-valid solves within  $2R_t$ . Every leave-one-domain-out  $D_3$  lies from +0.305 to +0.395, and every lower bound is at least +0.185. All of these results are descriptive.

## U.2 DAGGER ABLATION

The pre-registered ablation collects on-policy records with the seed-17 adapter under each additive algorithm, aggregates them with replay records from the frozen corpus, retrain, and evaluates on the validation panel and the held-out panel P2. The primary endpoint is the paired separation  $S$  of the DAGger adapter from the exact- $\epsilon$  0.75 rung pooled over the 23 tasks, with SEPARATED\_ABOVE requiring a positive lower bound. The co-primary is the change  $\Delta$  in M1 from the pre-DAGger adapter, with equivalence margin  $\pm 0.05$ .

An amendment at 11:21 UTC on 2026-09-24, decided before any DAGger training or evaluation outcome, cut the plan to seed 17 because the three-seed plan could not fit a hard wall-clock deadline. The running seed-29 collection was cancelled with 1.79 GPU-h still charged, and seed 71 was not run. Each algorithm’s training set became the first 512 on-policy seed-17 records plus 512 replay records, which the menu augmentation turns into 2048 samples over 64 steps, with the recipe otherwise unchanged.

All 24/24 validation-panel and 22/22 P2 DAGger episodes were independently replayed with no mismatch. The primary endpoint is NOT\_SEPARATED,  $S = +0.115 [-0.026, +0.258]$  over 23 tasks, and the co-primary is INCONCLUSIVE,  $\Delta = -0.030 [-0.114, +0.057]$ . As descriptive secondary results,  $S$  is +0.278 [+0.082, +0.473] on P2 and -0.034 [-0.197, +0.135] on the validation panel, and the last-label rate on the validation panel falls from 0.205 to 0.137. One small round at one seed therefore produces no detectable change in measured choice quality, since  $\Delta$  is inconclusive against the  $\pm 0.05$  margin, and the ablation is inconclusive.

## U.3 ZERO-SHOT POLICY

The second realisable policy is the pretrained Qwen3-VL-8B-Instruct base with no adapter, given the frozen scene-only choice-frontier observation, one appended system message stating the JSON format and asking for the frontier state whose scene appears closest to satisfying the goal pages, a fixed three-rule output extractor, greedy decoding at inference seed 17, and the adapter’s cap of

Table 22: Zero-shot base against random-valid ( $D_3$ ), the exact- $\epsilon$  0.75 rung ( $S$ ), and the three-seed adapter ( $A$ ), with paired task-clustered 95% intervals (10,000 draws, seed 133). Panel endpoints are pre-registered and pooled entries are descriptive. The  $p$  rows give exact sign-flip  $p$  values (descriptive). A  $p$  value tests a zero difference, and EQUIVALENT is the  $\pm 0.05$  margin rule.

|                | Validation (12, development) | P2 (11, confirmatory)   | Pooled (23, descriptive)    |
|----------------|------------------------------|-------------------------|-----------------------------|
| $D_3$          | +0.005 [−0.025, +0.044]      | +0.001 [−0.048, +0.051] | +0.003 [−0.024, +0.033]     |
| verdict        | EQUIVALENT                   | INCONCLUSIVE            | within margin (descriptive) |
| $p$            | 0.906                        | 1.000                   | 0.896                       |
| $S$ vs 0.75    | −0.321 [−0.422, −0.226]      | −0.188 [−0.239, −0.140] | −0.257 [−0.324, −0.197]     |
| verdict        | SEPARATED.BELOW              | SEPARATED.BELOW         | SEPARATED.BELOW             |
| $p$            | $4.88 \times 10^{-4}$        | $9.77 \times 10^{-4}$   | $2.38 \times 10^{-7}$       |
| $A$ vs adapter | −0.280 [−0.425, −0.142]      | −0.415 [−0.589, −0.248] | −0.344 [−0.459, −0.232]     |
| verdict        | SEPARATED.BELOW              | SEPARATED.BELOW         | SEPARATED.BELOW             |
| $p$            | 0.004                        | 0.004                   | $7.63 \times 10^{-6}$       |

Table 23: The seven surveyed published interfaces, with the survey’s pre-audit prediction for random-valid against the reference.

| Interface                            | Venue        | Type | CPU-feasible    | Prediction             |
|--------------------------------------|--------------|------|-----------------|------------------------|
| AutoPlanBench (Stein et al., 2025)   | ICAPS 2025   | B    | yes             | divergent              |
| ScienceWorld (Wang et al., 2022)     | EMNLP 2022   | B    | yes             | divergent              |
| GLAM (BabyAI-Text)                   | ICML 2023    | B    | yes             | divergent              |
| WebShop                              | NeurIPS 2022 | B    | no              | divergent              |
| RAP                                  | EMNLP 2023   | C    | no              | divergent              |
| LLM-First Search (Herr et al., 2025) | arXiv 2025   | C    | no <sup>†</sup> | saturated <sup>†</sup> |
| EmbodiedBench                        | ICML 2025    | B    | no              | divergent              |

<sup>†</sup> Audited later with a GPU reference model, where saturation under the paper’s budget follows by construction (LLM-First Search Audit appendix, Appendix V.1).

$2R_t$  calls. It ran on both additive algorithms over the validation panel and P2, 46 episodes in all. The preferred arm, InternVL3.5-8B trained with the frozen three-seed recipe, was ruled out before any outcome. Its mean observation takes 11,748 tokens against 3,111 for Qwen3-VL, which puts training alone at 43 to 55 GPU-h against the 30 GPU-h cap, and a reduced recipe would break the byte-identical condition, so it was not run in any form. The protocol was committed before any GPU launch, and no amendment followed.

Table 22 gives the endpoints.  $D_3$  names the pre-registered learned-minus-random-valid endpoint for both arms (a single realisation for the zero-shot arm). Validation-panel verdicts are development-stage and P2 verdicts are held-out confirmatory. On P2 the upper bound of  $D_3$  exceeds the  $\pm 0.05$  margin by 0.001, so the frozen rule returns INCONCLUSIVE rather than EQUIVALENT. The arm’s M1 is 0.026 [0.000, 0.063] on the validation panel, 0.017 [0.000, 0.051] on P2, and 0.022 [0.003, 0.046] pooled, below the exact- $\epsilon$  0.75 rung on every panel. All 1,592 model calls produced a valid choice through the first extractor rule. Heap-head agreement minus chance is +0.003 and +0.001, the last-label rate is 0.328 and 0.285 against a chance rate of about 0.13 to 0.14, and 4 of 46 episodes solve at  $2R_t$  (descriptive). The pre-registered prediction,  $D_3$  within the margin with  $S$  and the adapter contrast  $A$  separated below on both panels, held. All 46/46 episodes were independently replayed with no mismatch.

## V RESULTS DETAIL: EXTERNAL IDENTITY AUDIT

This appendix gives the interface survey and the ScienceWorld audit behind Section 5.5, with the LLM-First Search audit as a subsection.

The survey admitted published evaluations from 2022 on in which the model chooses from a runtime-supplied candidate set inside a planning or search loop, with public tasks and code. Seven candidates passed, five of type B (the model picks one valid action per step), two of type C (the model scores candidates that a search orders), and none of type A, our enumeration contract (Table 23). Four are not CPU-feasible under their papers’ budgets. For WebShop (free-text search, no step limit) and LLM-First Search (an unreported token cap) this was a judgement call, and LLM-First Search was the only saturated prediction, which the later audit in the LLM-First Search Audit appendix (Appendix V.1) tested. ScienceWorld is the most cited feasible candidate, and the runner-up shares its prediction.

Table 24: LLM-First Search success by configuration under the paper’s token budget and at the reference’s own number of expansions. Differences are reference minus random-valid with task-cluster 95% intervals (10,000 draws, seed 133, descriptive). The reference is the substituted local model, and its matched-budget success equals its token-budget success by construction. Over all 80 tasks, random-valid success at 1, 1.25, 1.5, 1.75, and 2 times the reference’s expansions is 0.0775, 0.0975, 0.14, 0.17, and 0.19 (normalised area 0.135).

| Configuration    | Reference | Random-valid | Paper token budget      | Random-valid | Matched decision budget |
|------------------|-----------|--------------|-------------------------|--------------|-------------------------|
|                  |           |              | Difference              |              | Difference              |
| Countdown 3 (20) | 1.000     | 1.000        | 0.000 [0.000, 0.000]    | 0.120        | +0.880 [+0.770, +0.970] |
| Countdown 5 (20) | 0.750     | 1.000        | −0.250 [−0.450, −0.100] | 0.110        | +0.640 [+0.370, +0.870] |
| Countdown 7 (20) | 0.650     | 1.000        | −0.350 [−0.550, −0.150] | 0.020        | +0.630 [+0.410, +0.840] |
| Sudoku 4×4 (20)  | 0.900     | 1.000        | −0.100 [−0.250, 0.000]  | 0.060        | +0.840 [+0.700, +0.950] |
| All (80)         | 0.825     | 1.000        | −0.175 [−0.263, −0.100] | 0.0775       | +0.747 [+0.648, +0.838] |
| Verdict          |           |              | SATURATED               |              | CHOICE_REGISTERED       |

The frozen protocol takes the first 200 test variations under the paper’s 100-step limit with five random-valid seeds (17, 5077, 6131, 7409, and 8527), and ten re-run episodes are byte-identical. The reference completes 183 of 200 (0.915), including 3/9 boil, 6/9 melt, 4/9 freeze, 6/9 change-state, 135/135 use-thermometer, and 29/29 measure-melting-point variations, and falls short only where 17 of the 36 Matter variations need more than 100 moves. Random-valid completes none, and 997 of its 1000 episodes end in a failure score.

The slice covers 6 of 30 task types (200 of 1,819 test variations), on all of which the paper’s own Random-Valid score is 0.00. The paper reports normalized scores, so its DRRN score of 0.084 on these tasks is not a success rate. The audit uses ScienceWorld 1.3.0, whose notes warn that gold paths can change relative to 1.2.x, and the paper’s simulator version was not re-run.

## V.1 LLM-FIRST SEARCH AUDIT

LLM-First Search is a type-C interface. The model scores candidates, and the search expands the argmax child or pops the highest-valued frontier node, so every unexpanded alternative stays in the frontier. Random-valid answers every query with independent uniform draws, spends zero tokens, and makes no model call, and a decision is one node expansion after the root. The upstream code (commit 3025bdaa, Apache-2.0) is imported unchanged and not vendored. The fixed rule audits every configuration whose published GPT-4o win rate exceeds 10%, which excludes Sudoku 6×6 and keeps Countdown 3, 5, and 7 and Sudoku 4×4, with games 0 to 19 of each, 80 tasks in all. Budgets are the paper’s per-game token caps (1,000,000 for Countdown and 100,000 for Sudoku 4×4). Four deviations are declared. The paper’s GPT-4o is replaced by the local Qwen3-30B-A3B-Instruct-2507, so reference success is that model’s, the reference runs once per task, Sudoku 6×6 is excluded, and random-valid uses a faster frontier implementation that produced byte-identical logs on 57/57 probe episodes. Random-valid uses the five seeds of the first audit.

All 80/80 reference and 400/400 random-valid episodes were independently replayed with no mismatch. No pair of the 400 is identical, and the first divergence has median index 0 and maximum 2. Table 24 gives both readings. Under the paper’s budget random-valid wins every episode at every seed, so the verdict is SATURATED, and every reference loss ends at the token cap. This outcome follows by construction. Random-valid spends no tokens and the frontier keeps every alternative, so it is a complete randomized search, and a pre-audit probe had won 100/100 non-audited episodes. Its only way to lose was a 500,000-expansion safety guard, which never bound. Random-valid used a median of 201 expansions and a maximum of 102,606, against a median of 29.5 and a maximum of 1,300 for the reference. Within the reference’s own number of expansions per task, random-valid wins 0.0775 (per seed 0.0625, 0.075, 0.075, 0.0875, and 0.0875), so the verdict is CHOICE\_REGISTERED. The reference’s matched-budget success equals its token-budget success by construction, so the budget curve in Table 24 is the evidence that choices matter. The matched-budget prediction held, and the paper-budget saturation follows by construction. Published GPT-4o success on these configurations is 0.768. One amendment, made before any audit episode, only reordered the smoke and audit start, and the ledger spent 2.89 of 12 GPU-h with no paid API use.

## W RESULTS DETAIL: SEED REPLICATION AND FOLLOW-UP LIMITS

This appendix reports the enumeration-contract seed replication and lists the limits of every follow-up window, for readers checking seed and panel scope.

The prospective training-seed replication (seeds 17/29/71 frozen pre-launch, 534 new episodes independently replayed) executed on two cells (Table 4). In the greedy  $\times$  multimodal replication cell (greedy best-first  $\times$  multimodal-state, 24-problem unseen panel), per-seed success is 21/24, 21/24, 22/24 (mean 0.889, range [0.875, 0.917]). The contrast against the pretrained base is positive at every seed (+0.875/+0.875/+0.917, no sign flip). The contrast against random-valid stays negative (−0.125/−0.125/−0.083). On the DAgger iteration-1 cell set (BFS  $\times$  three modalities, DAgger and continued-SFT arms), every arm and seed scores 0/72 unseen, with paired contrast +0.000 [+0.000, +0.000] at each seed. The published iteration-2 verdict (1/72 vs 1/72, seed 17) is unchanged. Both cells replicate across all three seeds. Cell A covers one of the three modalities of the greedy additive cell. The remaining two modalities and all other enumeration-contract cells use seed 17 only. Cross-seed aggregation is descriptive ( $n=3$ ). The image-only-arm proposal has since executed under the native-arms v2 amendment (the Image-Only Native Arms appendix, Appendix Q), and its v1 gate remains as history.

The follow-up windows share this development-stage status, and we list their limits separately. The native-arms window evaluates an 18-unit clean panel with one training seed per cell. The R1-R4 follow-up window reuses the frozen native-arms adapters for R1, R2, and R4, and trains one seed-17 run per cell for R3. The choice-frontier window trains one seed per algorithm on a 9-task panel whose discriminative subset holds one task. The comparator zoo is a CPU-only exploratory analysis. The adapter has three training seeds on the validation panel, the held-out panel P2, and the unscreened panel P2u, and the DAgger ablation trains one seed on half the planned corpus. None of these windows accessed the frozen 45-task final evaluation. The shared nine-task panel draws from 8 domains with 7 compact and 2 expanded variants.

**Final primary evaluation prerequisites.** No authorized protocol yet exists for the frozen 45-task final evaluation. Freezing one requires recorded decisions on the following prerequisites. (1) The condition-level data source and Search-Trace Segment construction with per-algorithm allocation. (2) Compute parity and checkpoint-selection rules. (3) The episode-budget unit, the whole-instance split procedure, and the fixed Search Memory capacity. (4) The estimand, the uncertainty interval or test, the success threshold, the invariant-compliance rule, and the validity and invalid-operation denominators for every cell. Until these choices are recorded, the final evaluation protocol cannot be frozen and no final comparative efficacy finding exists.

## X DECISION CONTRACTS

Table 25 compares the enumeration and choice-frontier contracts defined in Section 4.

## Y USE OF LLM STATEMENT

Large language model agents drafted and revised the manuscript text and wrote the figure scripts under the authors’ direction. An agent pass searched for citations, and a human verified every cited source. Agent reviewers ran repeated review rounds on the drafts. The authors verified every reported number against its pinned artifact and take full responsibility for the content.

Table 25: The two decision contracts. Rows compare what the policy emits, who owns the frontier, what the observation carries, how the budget binds, and the role of the identity audit. Results under the two contracts are not comparable as performance rows.

|                | Enumeration contract                                                                         | Choice-frontier contract                                                                                            |
|----------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Policy emits   | One Typed Search Operation per step, taken from a grounded, unscored candidate action menu   | One opaque label from a menu of frontier states, naming the state to expand next                                    |
| Frontier       | Kept in textual Search Memory with visited and score state, advanced by submitted operations | Owned by the runtime. Menu labels are re-assigned per decision by a seeded permutation                              |
| Observation    | Set by the Modality Observation adapter                                                      | Visual only, with task pages, the initial scene, one unlabelled 128px scene per frontier state, and goal pages      |
| Budget         | Learned arms capped at $2\times$ the exact-reference decision count, base at one call        | Decisions and expansions capped at $2\times$ the additive exact-reference expansion count, goal selection uncharged |
| Identity audit | Diagnostic run on the expanded-baseline additive cells                                       | Pre-registered pass condition over 18 control pairs                                                                 |